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REVIEW ARTICLE

Lichen Planus: What is New in Diagnosis and Treatment?

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Abstract

Lichen planus (LP), an idiopathic, multifaceted chronic inflammatory disease with a heterogeneous clinical presentation, affects approximately 0.5–1% of the population. The various clinical manifestations of LP fall into three broad categories, namely cutaneous, appendageal, and mucosal, with further subclassification depending on the morphology and distribution patterns of individual lesions. There is mounting evidence that LP has systemic associations, including autoimmune conditions, glucose intolerance, dyslipidemia, and cardiovascular disorders. Cutaneous hypertrophic and mucosal forms of LP are at a heightened risk for malignant transformation. Familiarity with these potential associations in conjunction with long-term follow-up and regular screening could lead to a timely diagnosis and management of concomitant conditions. In addition, the frequent quality of life (QoL) impairment in LP underscores the need for a comprehensive approach including psychological evaluation and support. Several treatment strategies have been attempted, though most of them have not been adopted in clinical practice because of suboptimal benefit-to-risk ratios or lack of evidence. More recent studies toward pathogenesis-driven treatments have identified Janus kinase inhibitors such as tofacitinib, phosphodiesterase-4 inhibitors such as apremilast, and biologics targeting the interleukin-23/interleukin-17 pathway as novel therapeutic options, resulting in a dramatic change of the treatment landscape of LP. This contemporary review focuses on the diagnosis and management of LP, and places emphasis on more recently described targeted treatment options.

Key Points

Lichen planus (LP), an idiopathic inflammatory disease with a heterogeneous clinical presentation, affects approximately 0.5–1% of the population.

There is mounting evidence that LP is not an isolated skin disorder, given the reports associating LP with autoimmune conditions, glucose intolerance, dyslipidemia, and cardiovascular disorders.

More recent research toward pathogenesis-driven treatments has identified Janus kinase inhibitors, phosphodiesterase-4 inhibitors, and biologics targeting the interleukin-23/interleukin-17 pathway as novel therapeutic options, resulting in a dramatic change of the treatment landscape of LP.

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1 Introduction

Lichen planus (LP), first described by Erasmus Wilson in the nineteenth century, is a multifaceted chronic inflammatory disease with diverse clinical manifestations affecting the skin, hair, nails, and mucosal membranes [1–3]. Its epidemiology and pathogenesis are incompletely understood [4]. Lichen planus can give rise to substantial morbidity and discomfort [4]. Certain infections, such as hepatitis C virus (HCV) and coronavirus disease 2019 (COVID-19), have been associated with LP [3, 5, 6], as have vaccines and medications [6–11]. Lichenoid drug eruptions (LDEs) can present with features indistinguishable from classic LP, emphasizing the importance of a thorough assessment of potential drug etiology in all patients presenting with a new diagnosis of LP [2]. The various clinical manifestations of LP fall into three general categories, namely cutaneous, appendageal, and mucosal, with further subclassification depending on the morphology and distribution patterns of individual lesions [4, 12, 13]. In the natural course of the disease, cutaneous LP resolves spontaneously within 6–18 months in most patients. However, hypertrophic cutaneous and mucosal variants frequently persist longer [3, 14, 15]. Several treatment strategies have been attempted, though most of

them have not progressed to clinical practice because of suboptimal benefit-to-risk ratios or lack of evidence [16]. More recent studies toward pathogenesis-driven targeted therapies have identified Janus kinase (JAK) inhibitors, phosphodiesterase-4 (PDE-4) inhibitors, and biologics as novel therapeutic options, resulting in a dramatic change of the treatment landscape of LP [4].

This contemporary review focuses on the diagnosis and management of LP, with emphasis on more recently described targeted treatment options.

2 Background

2.1 Epidemiology

The estimated incidence of LP varies depending on geographical location, ranging from 14 to 250 cases/100,000 person-years [4, 17–19]. In a cross-sectional study of the All of Us program from the USA, the overall prevalence of LP was 0.39% [20]. The incidence of LP is higher in dermatology and oral medicine clinics (1.5% and 5%, respectively) [19]. Regarding oral LP (OLP), a recent meta-analysis found a global pooled prevalence of 1.01%, with a marked geographical heterogeneity. Higher prevalences of OLP were reported from South America, Africa, and Europe [21]. For cutaneous LP, the mean age of onset has been classically described as the fourth and fifth decades of life, whereas it is approximately 10 years later for OLP (between 57 and 61 years) [22, 23]. Interestingly, however, in the All of Us database cross-sectional study from 2022, the average age of LP was 67 years and the prevalence increased with age [20]. Similarly, a meta-analysis noted an increasing prevalence of OLP after 40 years of age [21], with both studies highlighting the burden of this condition in older adults. Most studies identified no sex predilection in the cutaneous form, while OLP is up to three to four times more common among females [3, 4, 22, 24, 25].

There is scarce literature regarding the epidemiology of the appendageal variants of LP [26]. The overall prevalence of lichen planopilaris (LPP) was 0.043% in the All of Us database [20], while a review of the health records from New York found an overall prevalence of 0.032% for LPP and frontal fibrosing alopecia (FFA) combined [27]. In both studies, women over the age of 50 years accounted for the majority of the study population [20, 27]. There are, however, increasing descriptions of LPP and FFA occurring in men [26, 28]. Of note, the incidence of FFA has increased within the past two decades, suggesting a role for environmental influences in its etiology [29–32].

Pediatric LP represents approximately 1–4% of all cases [33, 34], with this figure being 18.7% in one large series from India [35]. Most reports of childhood LP are from the Indian subcontinent [13, 33, 36]. A systematic review on pediatric LP identified a slight male predominance [33]. It typically manifests in adolescence with symptomatic cutaneous involvement [37]. Compared with adults, nail involvement and linear LP appear to be more common in children [33, 38, 39], whereas oral mucosa involvement is less common in pediatric LP [33, 37]. Several LP variants have been reported in children, such as LP pigmentosus [40], atrophic form [37], perforating variant [41], LPP [33, 34, 42], and ocular LP [34].

2.2 Pathogenesis

The pathogenesis of LP is complex and incompletely elucidated, with a combination of numerous factors, such as genetic predisposition, environmental conditions, immune dysregulation, and exposure to medications or infectious agents ultimately leading to T-cell-mediated damage of basal keratinocytes [4, 39, 43, 44].

2.2.1 Genetics

Familial occurrence of LP has been described in up to 10.7% of the cases, with the familial form being associated with an earlier onset and generalized presentation with more frequent relapses compared with nonfamilial cases [14, 45, 46]. In addition, LP has been reported in monozygotic twins [39, 46, 47]. Several human leukocyte antigen (HLA) alleles have been associated with LP, such as HLA-B7 [39], DQB1*0201 (vulvovaginal gingival syndrome [48]), DR6 (HCV-associated OLP [49]), DRB1*11/DQB1*03 (LPP [50]), and HLA DQB1*05 [51]. Furthermore, phenome-wide association and case–control studies revealed single-nucleotide polymorphisms in pro-inflammatory cytokine and major histocompatibility complex regions, further supporting a genetic predisposition in LP [51–53].

2.2.2 Immunopathogenesis

Lichen planus is considered an autoimmune process, where the lesional and skin-infiltrating CD8-positive T cells recognize an exogenous or altered self-antigen associated with MHC class I on keratinocytes and cause epithelial damage [54–57]. Increased expression of intercellular adhesion molecule-1 by basal keratinocytes was demonstrated to enhance the interaction between the T cells and keratinocytes [13, 58]. Plasmacytoid dendritic cells play a central role in the pathogenesis by presenting the antigens and enhancing the inflammatory cascade via production of factors such as interferons (IFNs) [57, 59]. In turn, IFNs induce the expression of numerous cytokines and chemokines, creating a pro-inflammatory milieu. The cytokine profile

includes interleukin (IL)-9, IL17, IL-22, and IL-23, among others [4, 60, 61]. The interplay between the IFN-inducible chemokines, CXCL9 and CXCL10, and their receptor, CXCR3, further contributes to the pathogenesis [59]. Apoptosis of keratinocytes occurs through a number of different pathways [Fas ligand, granzyme-B-perforin, tumor necrosis factor (TNF)-α] and/or generation of reactive oxygen species, resulting in the formation of colloid bodies [39, 55, 57, 60, 62]. In addition, matrix metalloproteinases (MMP) such as MMP-9 are upregulated, which are thought to lead to basement membrane disruption [56, 63]. Different predominant T-cell subsets were detected in cutaneous LP and OLP ($CD4^+$ and $CD8^+$, respectively) [60]. A Th17 phenotype was associated with erosive OLP, whereas a skew toward Th2 and Treg phenotypes was found in the reticular form of OLP [57]. Regulatory T cells were detected in the circulation of patients with LP, showing that the inflammatory response is not restricted to the skin [64]. An excellent schematic illustration of LP immunopathogenesis is available in the review article on oral LP recently published in the *American Journal of Clinical Dermatology* (Fig. 1; Louisy et al.[57]).

In addition to T lymphocytes, mast cells were also shown to participate in disease pathogenesis and show signs of activation [4, 31, 65, 66]. Additional factors implicated in the LP pathogenesis are microRNAs [44, 61], sweat leakage into the dermal–epidermal junction (DEJ) [67], dysregulation of Notch signaling [68], abnormal Toll-like receptor activity [69], inflammasomes [70, 71], increased angiogenesis [13, 72], and expression of cathepsin K [73]. Regarding the pathogenesis of LPP and FFA, these entities are thought to have an autoimmune etiology with IFN-mediated collapse of the hair follicle bulge immune privilege and subsequent destruction of stem cells [31, 74]. Additional pathogenetic mechanisms include decreased peroxisome proliferator-activated receptor-γ signaling [75, 76], disturbance of the mTOR signaling [30], activation of the IL-23/IL-17 axis [66], epithelial-to-mesenchymal transition [31, 77], and neurogenic inflammation [32]. In addition, a study on 22 patients with LP found that IL-31 was overexpressed in LP lesions; however, the level of expression did not show correlation with intensity of pruritus in the same study [78]. The precise role of IL-31 as an itch mediator in LP remains to be elucidated.

2.2.2.1 *Janus* Kinase (JAK)–Signal Transducer and Activator of Transcription (STAT) Signaling Pathway in the Pathogenesis of Lichen Planus

Several studies have found the JAK–STAT pathway to be a cornerstone of LP pathogenesis [31, 79, 80]. An immunohistochemistry-based study showed STAT1 protein expression in the dermal infiltrate of cutaneous LP [80]. A transcriptomic profiling study demonstrated the importance of JAK2/STAT1 signaling in the LP inflammatory response, specifically playing a role in enhancing the sensitivity of the keratinocytes following IFN-γ stimulation [79, 81]. A further study highlighted upregulation of JAK1 and JAK3 in LP [82].

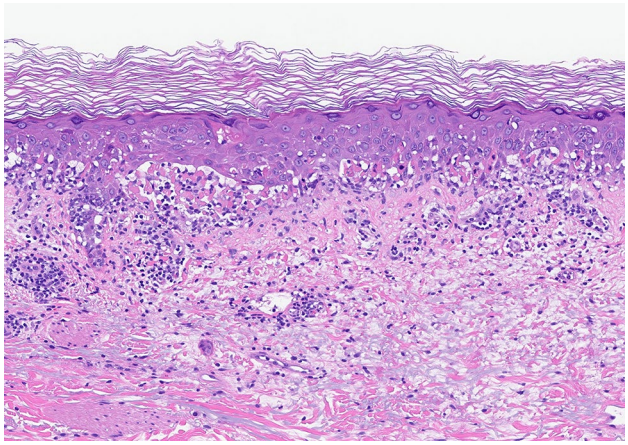


Fig. 1 Lichenoid reaction to immune checkpoint inhibitor therapy (nivolumab). Histopathology demonstrates lichenoid dermatitis with numerous apoptotic keratinocytes (H&E ×200)

2.2.3 Risk Factors, Potential Triggers, and Etiologic Associations

Several risk factors and triggers have been associated with LP, including medications, infections, contact-sensitivity reactions, smoking, mechanical trauma, and radiation [8, 15]. In addition, seasonal changes may influence the incidence of LP [83, 84]. Medications most frequently implicated in LDEs include beta-blockers [85], antimalarials, epidermal growth factor receptor-tyrosine kinase inhibitors [86], transforming growth factor beta inhibitors [87], thiazide diuretics, angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers [88], nonsteroidal anti-inflammatory agents, antidepressants, antidiabetics such as sitagliptin [89], gold salts, methyldopa, quinine [90], penicillamine, aminoglycoside antibiotics, proton pump inhibitors, denosumab, and TNF-α inhibitors [4, 88, 91–93]. There are many reports of LP occurrence with anti–PD-1 therapy [94–99] (Fig. 1). In particular, lichen planus pemphigoides (LPPemph) induced by PD-1 and PD-L1 inhibitors have garnered attention in the literature lately [100–103]. It has been hypothesized that the increased cytotoxic $CD8^+$ T cells from PD-1 inhibition may cause apoptosis of the basal keratinocytes and expose antigens of the DEJ to autoreactive T cells [102]. Notably, LPPemph may manifest several weeks after the cessation of immune checkpoint inhibitor therapy [102].

Among infectious agents, hepatitis C virus (HCV) has been most studied in the context of LP. A reciprocal association between HCV infection and LP was demonstrated, wherein both diseases appear to act as aggravating factors for one another [104]. The relationship is especially prevalent

in Mediterranean countries, Japan, and the USA; this geographical variability was attributed to differences in genetic susceptibilities [23, 53, 105]. A range of responses in LP was described following HCV treatment with interferon or direct-acting antivirals [106–108]. There is a growing body of evidence that hepatitis B virus (HBV) infection and liver cirrhosis may also be more common in patients with LP [109–111]. Regarding COVID-19, new onset or exacerbation of LP has been described following severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) vaccination or, less commonly, infection, following a mean latency period of 11–13 days [5, 6, 10, 112]. In addition to SARS-CoV-2 vaccines, the development of lichenoid reactions was also reported following HBV [7], influenza [113], rabies [8], and tetanus–diphtheria–acellular pertussis vaccinations [11]. Additional microorganisms that may be involved in LP pathogenesis and/or act as aggravating factors include *Candida albicans* [114, 115], human papilloma virus [57, 116, 117], human herpes virus 7 [118, 119], and periodontal pathogens such as *Aggregatibacter actinomycetemcomitans* [120, 121].

The association between OLP and contact sensitivities to metals (such as amalgam or copper), food flavorings, and plastics has been well established [4, 13, 38, 122, 123]. Furthermore, removal of the sensitizing metal has been documented to lead to improvement [122]. Based on these findings, patch testing has been recommended as a component of the evaluation of OLP by some authors [122, 124]. There are also anecdotal reports describing contact hypersensitivities in patients with cutaneous LP [40, 125, 126] and LPP/FFA [29]. Koebnerization has also been observed in various LP forms, including papular LP [127], linear LP [88], LPP/FFA (following hair transplantation or face-lift surgery) [38, 128, 129], and OLP (e.g., following trauma from sharp cusps) [130]. Radiotherapy-associated LP (with some cases reported in the context of combination therapy with nivolumab [94, 98]) may occur as either new-onset dermatosis or reactivation of pre-existing lesions. Although it typically occurs at the site of radiation, lesions may also become widespread [131–133].

3 Clinical Features and Variants

3.1 Cutaneous Lichen Planus

The typical presentation of cutaneous LP is with flat-topped, pruritic papules and plaques of varying size, covered by a white reticular scale called Wickham striae (WS), preferentially involving the wrists, forearms, and dorsal feet [2, 3, 16, 134]. Lesions become more generalized in approximately one-third of the patients [3]. In fair skin, the individual plaques are described as violaceous to reddish purple,



Fig. 2 Annular lichen planus on the lower leg in a patient with skin of color

though this color change is less prominent in patients with skin of color. In addition, postinflammatory pigmentary changes are more common in dark-complexioned individuals [135, 136] (Fig. 2). Many described subtypes deviate from the classical presentation and show variations by lesional morphology, distribution, configuration, or clinical course, as summarized below and in Table 1.

3.1.1 Cutaneous Lichen Planus Variants by Lesion Distribution

Several LP variants are delineated by a specific localization or distribution of lesions on the body, including actinic, inverse, linear, and palmoplantar subtypes (Table 1). Actinic LP, also known as LP subtropicus, mainly affects the sun-exposed areas in dark-complexioned individuals. It is thought to be precipitated by ultraviolet light exposure and has a high prevalence in India, Tunisia, and the Middle East [13, 16, 137, 138]. Inverse LP primarily involves the axillae, inguinal folds, and flexural surfaces of the extremities [16, 139]. Linear LP, also known as Blaschko-linear or Blaschkoid LP, is a rare variant typically manifesting with unilateral lesions along the lines of Blaschko, although bilateral involvement was rarely reported [140, 141] (Fig. 3). In facial linear LP, lesions may exhibit frequent folliculotropism, with a tendency for the development of atrophic lesions [140]. Linear LP shows marked clinical and immunophenotypical overlap with lichen striatus, posing a diagnostic challenge, especially in the pediatric age group. The presence of pruritus and mucosal involvement favor the former [142]. A subset of linear LP cases may have a (multi) dermatomal distribution, also referred to as zosteriform LP [143]. Palmoplantar involvement of LP is rare and differs from classical LP by the frequent absence of WS [144].

3.1.2 Cutaneous Lichen Planus Variants by Lesion Morphology

Hypertrophic LP (HLP) deserves special mention as a distinct LP variant characterized by larger, hyperkeratotic, and

Table 1 Lichen planus (LP) variants and subtypes covered in this review

LP variant/subtype	Salient clinicopathologic findings	References
Cutaneous LP		
Lesion distribution		
Actinic LP (LP subtropicus)	Hyperpigmented lesions with a predilection for sun-exposed areas	[137, 138]
Inverse LP	Predominant or exclusive involvement of intertriginous areas	[139]
Linear LP		
Blaschko-linear or Blaschkoid LP	Distribution of lesions along the lines of Blaschko	[126, 140, 141]
Zosteriform LP	Distribution of lesions along dermatome(s)	[143]
Palmoplantar LP	LP characterized by involvement of palms and soles	[144]
Lesion morphology		
Annular LP	Annular plaques with a predilection for the male genitalia and body folds	[156]
Atrophic LP	Progressively enlarging plaques with central atrophy, characterized by reduction of elastic fibers on histopathology	[147]
Annular atrophic LP	Violaceous plaques with an annular configuration and atrophic center	[146]
Erythrodermic LP	Generalized involvement by LP	[157]
Follicular LP	Follicular involvement by LP lesions	[4]
LP follicularis tumidus	Erythematous LP plaques accompanied by comedo-like and cystic lesions	[96]
Hypopigmented LP	Hypopigmented macules	[155]
Hypertrophic LP	Hyperkeratotic red-brown plaques and nodules with a variable verrucous morphology, with a predilection for the anterior legs Variably prominent epithelial hyperplasia accompanied by lichenoid inflammation with admixed eosinophils concentrated around the rete ridges	[43, 54]
"Invisible" LP	Pruritic lesions with a lichenoid histopathology; not readily discernible on clinical examination but may be recognized with the aid of Wood's lamp examination	[158]
LP pigmentosus	Insidious onset of gray to dark-brown macules and patches in sun-exposed and/or intertriginous surfaces (LP pigmentosus-inversus)	[149, 150]
Perforating LP	Keratotic papules and plaques showing transepidermal elimination on pathology	[41]
Porokeratotic LP	Lichenoid lesions with the additional finding of cornoid lamella on histopathologic examination	[160, 161]
Vesiculobullous LP	Vesicles and tense bullae located on pre-existing LP lesions	[97, 159]
LP overlap syndromes		
LP pemphigoides	Combines the features of LP and bullous pemphigoid. Typical presentation is with lichenoid plaques on extremities followed by an eruption of tense blisters	[163, 164, 167]
Lupus erythematosus/LP overlap	Combines the features of LP and lupus erythematosus	[169, 171]
Mucosal LP		
Oral LP		[21, 57, 175, 178]
Reticular	White, lace-like lesions	
Erosive	Persistent and painful erosions	
Erythematous (atrophic)	Clinically similar to the erosive form, except for more prominent atrophic lesions on a background of erythema	
Papular	Small, white papules	
Plaque-like	Homogeneous white plaques, more common among smokers	
Bullous	Blisters with a tendency to rupture	
Genital LP		
Female genital LP (vulvovaginal LP)		[48, 173, 179, 181, 183]
Erosive	Erythematous vulvovaginitis, ulcerated lesions, and scarring	
Vulvo-vaginal-gingival syndrome	Involvement of vulvar, vaginal, and gingival mucosae by LP	

Table 1 (continued)

LP variant/subtype	Salient clinicopathologic findings	References
Classic	Pruritic papules of variable color	
Hypertrophic	Thick violaceous plaques	
Male genital LP		[188, 189]
Annular	Ring-shaped plaques occurring via convergence of multiple papules	
Papular	Violaceous, flat-topped papules	
Less common forms reported such as atrophic, erosive, or bullous		
LP involving other mucosal sites		
Esophageal LP	Involvement of the upper and middle third of the esophagus by LP, most commonly associated with dysphagia	[190–192]
Laryngeal LP	A very rare presentation of LP, associated with hoarseness	[15]
Ocular LP	Typical presentation with cicatricial conjunctivitis, frequently accompanied by severe LP of other mucosal surfaces	[19, 193]
Otic LP	Can present with bilateral hearing loss, stenosis of the external ear canal, persistent otorrhea, tinnitus, and pruritus	[194, 195]
Anal LP	Involvement of the anal mucosa can lead to stenosis, pain, and constipation	[196]
Appendageal LP		
Lichen planopilaris		
"Classic" lichen planopilaris	Perifollicular erythema and hyperkeratosis with keratotic acuminate plugs, followed by destruction of hair follicles resulting in scarring alopecia	[74, 199, 200]
Graham–Little–Piccardi syndrome	Triad of cicatricial alopecia of the scalp, follicular lichenoid papules on the trunk and extremities, and nonscarring alopecia of axillary and pubic regions	[203]
Frontal fibrosing alopecia	Symmetrical and progressive loss of the scalp hair over the frontal hairline with recession accompanied by loss of eyebrows and eyelashes, thinning of axillary and pubic hair, yellow facial papules, and depression of the frontal veins	[28, 206, 208]
Nail LP		[39, 69, 213, 214, 217, 219, 220]
Nail matrix LP	Most common variant of nail LP; presents with nail plate thinning, longitudinal ridging and fissuring, with a potential for pterygium formation and anonychia	
Nail bed LP	Presents with onycholysis, nail bed atrophy, and subungual hyperkeratosis	
Trachyonychia	Thin and brittle nails, accompanied by diffuse longitudinal ridging and pitting	
Bullous/erosive LP	Manifests with nail bed erosions and formation of bullae with hemorrhagic crusts, which can culminate in permanent nail loss	
Idiopathic atrophy of the nails	Progressive and asymptomatic variant leading to thinning and atrophy of nails, more common in the pediatric age group	
Yellow nail syndrome-like LP	Toenail changes reminiscent of yellow nail syndrome (e.g., transverse over-curving, thickening, and yellow discoloration)	

scaly nodules and plaques, which can assume a verrucous appearance, showing a symmetric involvement of sun-exposed sites on the lower legs. The plaques are intensely pruritic and may be surrounded by numerous polygonal papules [38, 54]. In contrast to classic LP, HLP follows a more chronic course and may be associated with an increased risk of SCC development [43, 54], as further detailed in Section 4.2 below. Atrophic LP is another LP variant with a predilection for the legs. The lesions are characterized by papules that enlarge centrifugally and develop central atrophy, the histopathological correlate of which is diminished elastic fibers in the superficial dermis [16, 145–147]. A subset of cases of atrophic LP is thought to be the end-stage manifestation of hypertrophic LP [148].

LP pigmentosus manifests with gray to dark-brown macules and patches in photodistributed and/or

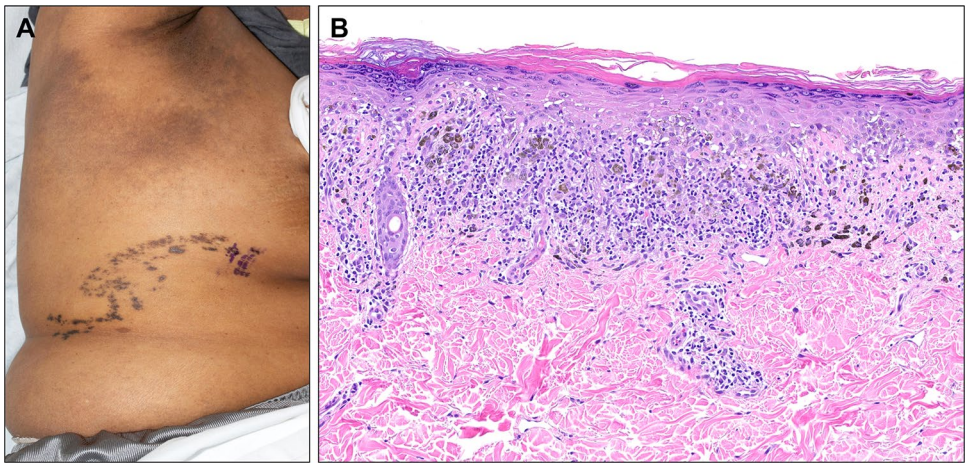


Fig. 3 **A** Blaschko-linear lichen planus on the trunk. **B** Interface dermatitis with numerous dermal melanophages (H&E ×200)

intertriginous areas (LP pigmentosus-inversus), possibly triggered by sunlight, friction, or exposure to certain chemicals such as mustard oil [149, 150] (Fig. 4A). The pigmentation pattern varies greatly and may be diffuse, follicular, reticular, ring-shaped, or blotchy [150]. Palmoplantar involvement may be more common than previously reported [151]. LP pigmentosus may be an abortive manifestation of LP, wherein the lichenoid inflammation quickly subsides, ensued by pigment incontinence which tends to be persistent [150] (Figs. 4B and 5). Several other entities, including erythema dyschromicum perstans and idiopathic eruptive macular pigmentation, share clinicopathologic features with LP pigmentosus; the overarching term acquired macular pigmentation of uncertain etiology was coined to encompass these conditions [152, 153]. LP pigmentosus has been more commonly described in patients with skin phototypes III–VI in India, Latin America, and the Middle East [150, 154]. Although LP pigmentosus is a well-recognized LP variant, hypopigmented LP is vanishingly rare [155]. Similarly, there are many other infrequently reported morphologic LP variants, such as annular [156], erythrodermic [157], follicular [96], "invisible" [158], (vesiculo)bullous [159], perforating [41], and porokeratotic variants [160, 161], which are outlined in Table 1.

3.1.3 Lichen Planus Overlap Syndromes

LPPemph is a distinct subepithelial blistering dermatosis combining the features of LP and bullous pemphigoid (BP) [162, 163]. It has been postulated that the lichenoid inflammation may expose basement membrane antigens, which subsequently promotes an autoimmune response ("epitope spreading" phenomenon). In keeping with this hypothesis, circulating autoantibodies against type XVII collagen (COL17, BPAG2; BP180) were demonstrated in patients with LPPemph. However, these target an epitope that is distinct from that typically implicated in the development of BP (i.e., NC16-A BP180) [164, 165]. This hypothesis is further supported by the clinical presentation of LPPemph, wherein the eruption of tense blisters is generally preceded by lichenoid plaques (Fig. 6). The classic areas of involvement are the extremities; involvement of the oral mucosa and vulva has also been reported [101, 164, 166]. Rarely, LPPemph may present as a toxic epidermal necrolysis-like eruption [165]. The term LPP pemphigoides was coined to describe a variant of LPPemph hallmarked by perifollicular separation and lichenoid inflammation [100]. Interestingly, LPPemph may rarely be devoid of blisters, an entity named "non-bullous LPPemph" [167]. An important differential diagnosis of LPPemph is bullous LP, which differs from the former in that the blisters form as a consequence of intense inflammation, without the presence of autoantibodies targeting type XVII collagen. In bullous LP, blisters are confined to the LP lesions, whereas they can also develop on previously unaffected skin in LPPemph [159, 162, 167]. Another important differential diagnosis is BP. Of note, LPPemph generally manifests in the third to fourth decades of life, earlier than the typical age of onset of BP [164, 168].

Lupus erythematosus (LE)/LP overlap syndrome combines the characteristics of LE and LP [13]. Direct immunofluorescence (DIF) studies play a vital role in the diagnosis of this cross-over syndrome by showing mixed features of LE (immunoglobulin and complement deposition in a granular pattern along the DEJ) and LP (e.g., shaggy fibrinogen deposition at the DEJ) [169–171]. LE/LP overlap syndrome may affect acral surfaces as a manifestation of chilblain LE [172].

3.2 Mucosal Lichen Planus

Virtually any mucosal surface can be involved by LP, which emphasizes the importance of multidisciplinary approach in the management of this protean condition [4]. The

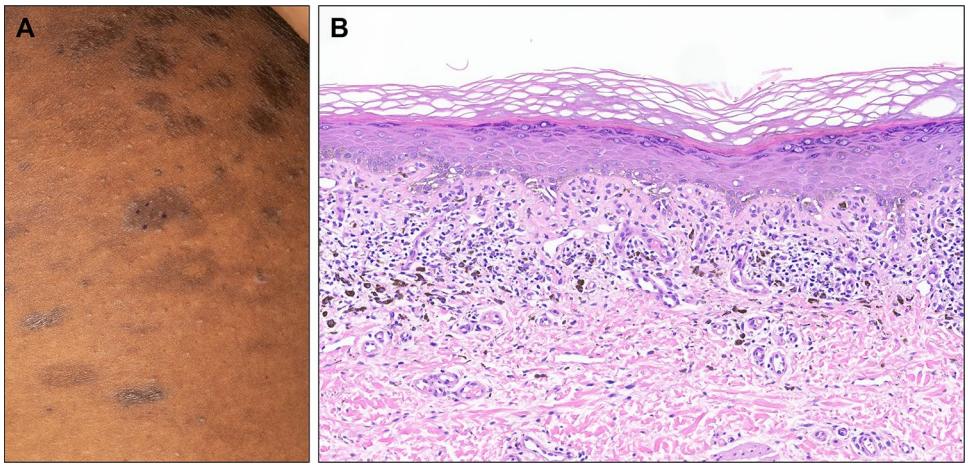


Fig. 4 **A** Lichen planus pigmentosus presenting with gray to dark-brown plaques. **B** Interface dermatitis with pigment incontinence (H&E ×200)

symptomatology is broad and depends on the affected site [15, 57]. In contrast to most variants of its cutaneous counterpart, mucosal LP tends to follow a protracted course with persistent or relapsing lesions [173].

3.2.1 Oral Lichen Planus

Oral LP is noted in up to 90% of patients with cutaneous LP [174, 175]. It is characterized by white, lace-like patches and plaques localized on bilateral buccal mucosal surfaces, followed by the tongue and gingiva, with a frequently bilateral distribution [57, 174]. Erythematous (atrophic) and erosive presentations are also relatively common and may be associated with symptoms such as burning, dysgeusia, or intense pain, in contrast to exclusively reticular OLP, which tends to be asymptomatic [21, 176]. Erosive oral LP manifests as desquamative gingivitis in approximately a quarter of patients [177]. Other less commonly encountered clinical forms of OLP include papular, plaque-like, and bullous [175, 178] (Table 1).

3.2.2 Genital Lichen Planus

3.2.2.1 Female Genital Lichen Planus Vulvovaginal LP (VVLP) is relatively common, present in approximately a quarter of women with mucosal LP [145, 173, 179]. The mean age of onset is in the fifth or sixth decade [180, 181]. Female genital LP is classified into three main categories, namely erosive, classic, and hypertrophic, with the erosive type being the most prevalent [181–183]. The most commonly involved sites are labia minora and clitoris [184]. Vaginal involvement is present in the majority of patients and appears to be underreported in the literature [173]. Gynecological examination is recommended to evaluate vaginal involvement in women with genital LP [184]. Lesional morphology of VVLP spans a broad spectrum from glazed erythema to ulcerations and scarring resulting in introital stenosis, urinary outflow obstruction, clitoral phimosis and burying, and vaginal adhesions and obliteration [38, 173, 180, 181, 185]. A large cohort study found vulval scarring in nearly all women with erosive VVLP, highlighting the destructive nature of this condition [179]. Oral LP is present in approximately half of the patients with LP of the vulva [179]. Vulvo-vaginal-gingival syndrome (VVGS) is a severe subtype with a propensity for scarring, characterized by erosive lesions involving the vulvar, vaginal, and gingival mucosae. Not all patients, however, have concomitant involvement of these three anatomic regions [179, 181]. Of note, strictures may be noted in other sites, as well, such as the esophagus and/or lachrymal ducts [48]. The coexistence of vulvo-vaginal-gingival syndrome and LPP was termed "vulvovaginal-gingival-pilar syndrome" [186].

3.2.3 Male Genital Lichen Planus

The most frequent morphology of male genital LP is annular, followed by papular. Albeit less common, other clinical forms such as atrophic, erosive, or bullous were also described [16, 187]. Glans penis and coronal sulcus are most commonly affected [187, 188]. In a study of 89 patients with

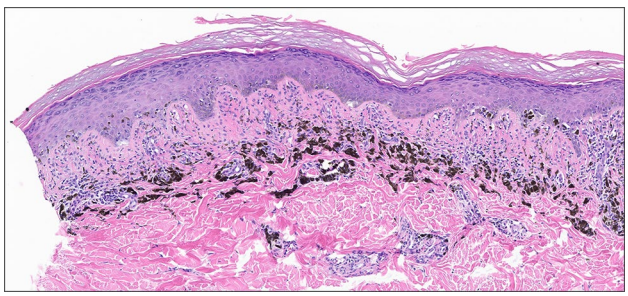


Fig. 5 In this case of lichen planus pigmentosus, the interface dermatitis is less prominent, whereas the pigment incontinence is the main histopathologic feature (H&E ×100)

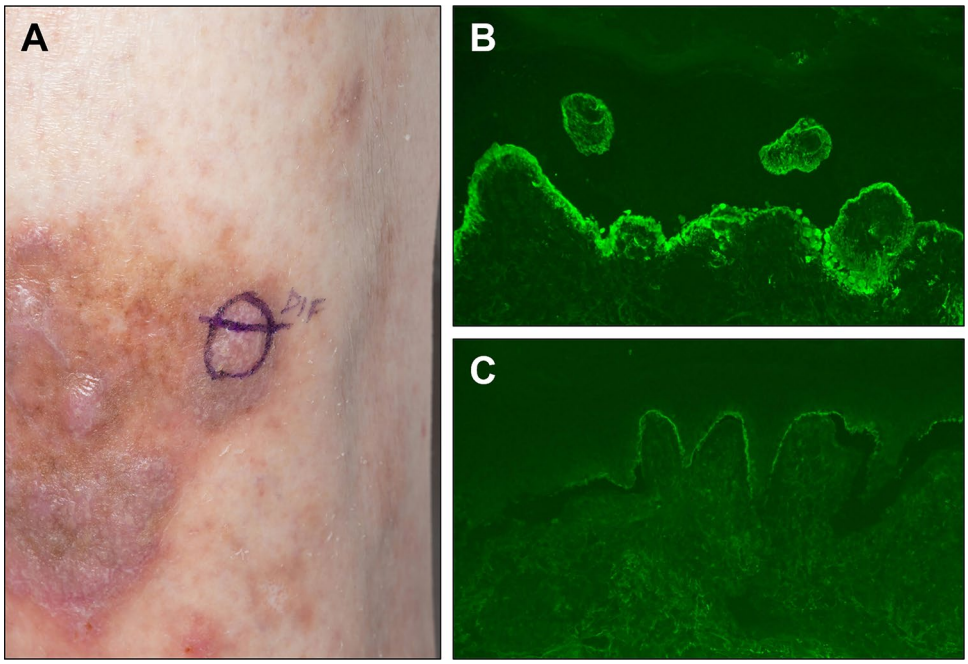


Fig. 6 Lichen planus pemphigoides. **A** Deflated blisters on a background of lichenoid plaque. **B** Fibrinogen deposition at the basement membrane zone on direct immunofluorescence examination from a lesional biopsy. **C** Deposition of IgG at the basement membrane zone on direct immunofluorescence examination from a perilesional biopsy

male genital LP, more than half of the patients were asymptomatic [189]. Nearly all patients in the same study were uncircumcised [189], whereas a meta-analysis suggested that genital LP was more common in circumcised men [148].

3.2.4 Lichen Planus of Other Mucosal Sites

Esophageal LP is a possibly underreported condition that can cause dysphagia, odynophagia, hoarseness, weight loss, and hematemesis [190–192]. Esophago-gastro-duodenoscopy is recommended in patients with LP who present with these symptoms [190, 192]. Esophageal LP most commonly involves the upper and middle thirds of the esophagus in middle-aged women [191]. Left untreated, scarring strictures can result [192]. Among patients with LP, the differential diagnosis of hoarseness also includes laryngeal LP, which is vanishingly rare [15, 38]. Ocular LP typically manifests with cicatricial conjunctivitis and is commonly accompanied by severe involvement of other mucosal sites. Corneal ulceration and visual impairment may develop as possible sequelae [19, 193]. Lacrimal canal pathologies, dry eye, and epiphora were also described in the context of LP [193]. Otic LP can give rise to hearing loss, which is usually bilateral and can have a conductive, sensorineural, or mixed pattern. Other manifestations within the spectrum of otic LP include otorrhea, tinnitus, plugging, and pruritus [194, 195]. The anal mucosa may also be involved by LP, potentially resulting in anal stenosis, constipation, and pain [57, 196].

3.3 Appendageal Lichen Planus

3.3.1 Lichen Planopilaris

Lichen planopilaris is the leading cause of lymphocyte-mediated cicatricial alopecia and end-stage pseudopelade of Brocq [197]. The majority of patients are postmenopausal women [198], although LPP is increasingly being recognized among men [26, 199]. Among women with LPP, hormone replacement therapy and surgically induced menopause were described in nearly one-fifth of patients, highlighting the role of hormonal factors in the etiology [74]. Patients manifest with a constellation of symptoms including hair thinning, pruritus, scalp tenderness, and/or burning [200]. Early lesions are characterized by perifollicular erythema and scale accompanied by keratotic acuminate plugs. Destruction of hair follicles leads to more established lesions with scarring and sparse "lone hairs" [2, 4, 200, 201]. Three distinct clinical variants have been recognized, namely the "classic" LPP, Graham–Little–Piccardi syndrome (GLPS), and FFA [128, 201, 202]. In the active stage of classic LPP, there are multiple, expanding alopecic patches in the frontal-central scalp and vertex, which are surrounded by grouped follicular papules [2]. Genital or extremity hair loss may be present alongside [74]. Involvement of other sites by LP is observed in 16.4–50% of patients with LPP [13, 74]. The triad of multifocal cicatricial alopecia involving the scalp, follicular lichenoid papules on the trunk and extremities, and noncicatricial hair loss in the axillary and pubic regions is termed GLPS [2, 203, 204].

Progressive and symmetrical loss of the scalp hair over the frontal hairline with recession is characteristic of FFA [205, 206]. Other manifestations of FFA include yellow facial papules corresponding to hypertrophic sebaceous glands, depression of the frontal veins, loss of eyebrows and eyelashes, glabellar red dots, hypopigmentation of the frontal hairline, loss of sideburns, and thinning of axillary and pubic hair [205–209]. In fact, eyebrow loss was reported in up to 60.1% of the patients [208], while 58.3% of the patients had decrease in axillary hair density [205]. Some authors argued that the nomenclature of FFA may be a misnomer, as this name restricts a condition with varied manifestations to a limited body region [14]. A link between long-term use of facial sunscreens/moisturizers and higher likelihood of FFA was suggested by some studies [210, 211]; however, this association was not reproduced in all cohorts [209, 212].



Fig. 7 Lichen planus involving the fingernails with periungual inflammation and dystrophic nail including pterygium formation

3.3.2 Nail Lichen Planus

Nail LP is present in 10–28% patients who have cutaneous LP [69, 213]. Nail LP is most frequently associated with OLP [39]. Fingernails are affected more commonly compared with toenails [69, 213–215] (Fig. 7). Involvement of the nail matrix leads to lateral nail plate thinning with longitudinal ridging and splitting, fragmentation of the lamina, and erythematous lunula [69, 213, 214]. In up to 20% of patients, dorsal pterygium formation occurs as a consequence of scarring damage to the germinative component of the nail matrix [39, 216, 217], with anonychia developing as a sequela in a subset of patients [217]. It is important to note that untreated or inadequately treated nail LP has the potential to show rapid exacerbation and result in permanent nail loss [39, 69, 215]. In contrast to LP of the nail matrix, LP of the nail bed is relatively infrequent and manifests with a variable combination of onycholysis, nail bed atrophy and erythema, subungual hyperkeratosis, and splinter hemorrhages [214, 216–218]. Isolated nail bed LP is very rare and, if present, usually affects the toenails [39, 69, 214]. Involvement of the nailfold, such as a ragged cuticle, has also been reported [214]. Other manifestations of nail LP include trachyonychia, bullous/erosive LP, idiopathic atrophy of the nails, and yellow nail syndrome-like LP [39, 214, 219, 220] (Table 1).

4 Associated Conditions

4.1 Autoimmune and Other Systemic Disorders

There is a growing body of evidence that LP is not an isolated cutaneous disease. Understanding the scope of the associated systemic disorders can offer an opportunity for a timely diagnosis and treatment [221]. A well-established link exists between LP and autoimmune/immune-mediated conditions, such as alopecia areata, autoimmune thyroiditis, celiac disease, dermatomyositis, inflammatory bowel disease/ulcerative colitis, LE, Sjögren syndrome, vitiligo, morphea, and myasthenia gravis [4, 221–223]. There have been conflicting results regarding the odds of rheumatoid arthritis and type 1 diabetes mellitus (DM) among patients with LP [221, 223], with some studies reporting an increased incidence [223, 224]. Another entity that appears to be overrepresented in patients with LP is thymoma, most commonly type A or AB [225, 226]. Thymectomy was not shown to improve the clinical course of LP in most instances [13, 225, 227]. Thymoma with immunodeficiency, also referred to as Good syndrome, is another condition anecdotally associated with LP [227–229].

Several studies pointed toward an association between LP and cardiovascular (CV) disease or risk factors [230–233]. Markers of subclinical atherosclerosis, such as carotid intima media wall thickness and epicardial fat tissue, were shown to be increased in patients with LP, even in the absence of clinically evident CV disease [230, 231, 234]. In addition, LP appears to be associated with higher abdominal circumference and blood pressures [230, 235]. Regarding biochemical parameters, patients with LP were shown to have levels of higher fasting blood glucose, homocysteine, C-reactive protein, erythrocyte sedimentation rate, and fibrinogen compared to controls [230, 235, 236]. In addition, LP is associated with higher fasting blood glucose levels, glucose intolerance, and increased rates of type 2 DM [235, 237, 238]. Dyslipidemia was also found to be more common among patients with LP, highlighting the importance of lipid level screening in this patient population [230, 233, 238, 239]. Taken together, LP is increasingly being regarded as a marker for metabolic syndrome [235, 236, 240], with longer disease duration and mucosal disease being additional possible risk factors [235]. From a pathogenesis standpoint, the

pro-inflammatory cytokines and chronic inflammation may interfere with insulin signaling and metabolism of carbohydrates and lipids [236, 241].

Similar to cutaneous and mucosal variants of LP, the appendageal variants have also been linked to a number of comorbidities [198]. Hypothyroidism is present in up to one-third of patients with LPP [74]. Additional conditions reported to be overrepresented in LPP include hirsutism, rosacea, skin cancer [198], and systemic LE (in females) [242]. Hyperlipidemia, on the other hand, does not appear to be more common in LPP compared with controls, suggesting that the immunologic mechanisms slightly differ between LPP and cutaneous/mucosal LP [242–244]. In the context of FFA, the frequency of thyroid dysfunction and atopy was found to be elevated [208].

4.2 Lichen Planus and Malignancy

A limited number of epidemiologic studies investigated the relationship between LP, internal malignancy, and associated mortality [22, 245]. In a large-scale study based on Finnish registry data, women with LP had increased mortality from non-Hodgkin and Hodgkin lymphomas and cancer of the oral cavity and tongue in comparison with the general female population [22]. A registry-based study from Sweden also noted an increased risk of oral cancer but did not identify an increase in internal malignancy [245].

There is a well-established risk of malignant transformation in mucosal LP forms [21, 246–250]. The World Health Organization (WHO) classifies OLP as a potentially malignant disorder [251]. In a meta-analysis summarizing 82 studies, OLP was found to confer a 1.14% risk of oral squamous cell carcinoma (SCC) development [247]. In a 25-year study based on patients with OLP from the population of Olmsted County, Minnesota, the incidence of oral SCC development was 3.1% [252]. Of note, oral lichenoid reactions and lesions also confer an increased risk for transformation [247], and some authors argue the malignant transformation risk is lower at 0.44% if strict clinical and histopathological diagnostic criteria are used for OLP [253]. Risk factors for malignant transformation include infection with HCV and *Candida albicans*, female gender, excessive alcohol consumption, smoking, and localization on the tongue. In addition, the propensity is substantially higher in atrophic/erosive LP compared with exclusively reticular OLP [21, 57, 115, 175, 247, 249, 250]. Oral LP-derived SCC appears to differ from conventional oral SCC in that the former tends to be multiple, well differentiated, early stage, and associated with a lower mortality rate (5-year mortality of 15.5% versus 35–50%, respectively) [57, 246, 248, 250]. The molecular mechanisms implicated in malignant transformation of oral LP are heterogeneous and include changes involving the apoptosis and DNA damage response pathways and epithelial–mesenchymal transition, among others [248, 254, 255]. It is recommended that patients with OLP undergo regular surveillance for the development of SCC [175, 246]. In a study including 303 patients with OLP, the median duration from the diagnosis of OLP to development of oral SCC was 2.9 years, emphasizing the importance of long-term follow-up [249].

Regarding genital LP, dysplastic transformation is estimated to develop in 2–3% of patients with vulvovaginal LP [13, 179, 181, 182, 185, 256]. It has been suggested that vulvar SCCs arising in association with LP may have a high frequency of inguinal metastases and rate of death due to disease (approximately 30%) [256]. A large population-based study identified LP as a significant risk factor for penile intraepithelial neoplasia [257]. Case reports of SCC arising in association with penile LP further reiterate the risk of progression in this condition [258, 259]. With regard to esophageal LP, the risk of SCC development has been estimated around 6% [192, 260].

Among the cutaneous forms of LP, hypertrophic LP stands out due its higher propensity for malignant transformation. In fact, more than half of the SCCs developing in the context of cutaneous LP are thought to arise in association with the hypertrophic variant [261]. Clinically, SCC in the setting of hypertrophic LP more commonly develops on the lower extremities of women [43]. It has been suggested that these carcinomas tend to be well differentiated and may have favorable outcomes [43, 261, 262] (Fig. 8). The duration between the initial occurrence of hypertrophic LP and development of SCC can be longer than a decade, reiterating the importance of long-term screening [261]. Of note, in addition to being a risk factor for malignant transformation, hypertrophic LP may also mimic SCC clinically and histopathologically (particularly in superficial biopsies) due to the presence of marked epidermal hyperplasia, posing a diagnostic challenge [43, 54, 263]. Development of SCC in association with nail LP has also been described, albeit very rarely [39, 69, 264].

5 Diagnosis

The diagnosis of LP is rendered through a combination of clinical history, physical examination, and histopathologic evaluation. Direct immunofluorescence testing (DIF) is an important adjunct diagnostic modality [13].

5.1 Diagnosis and Differential Diagnosis of Cutaneous Lichen Planus

As the prototype of the lichenoid reaction pattern, LP is histopathologically characterized by a band-like lymphocytic inflammation obscuring the DEJ and resulting in pointed

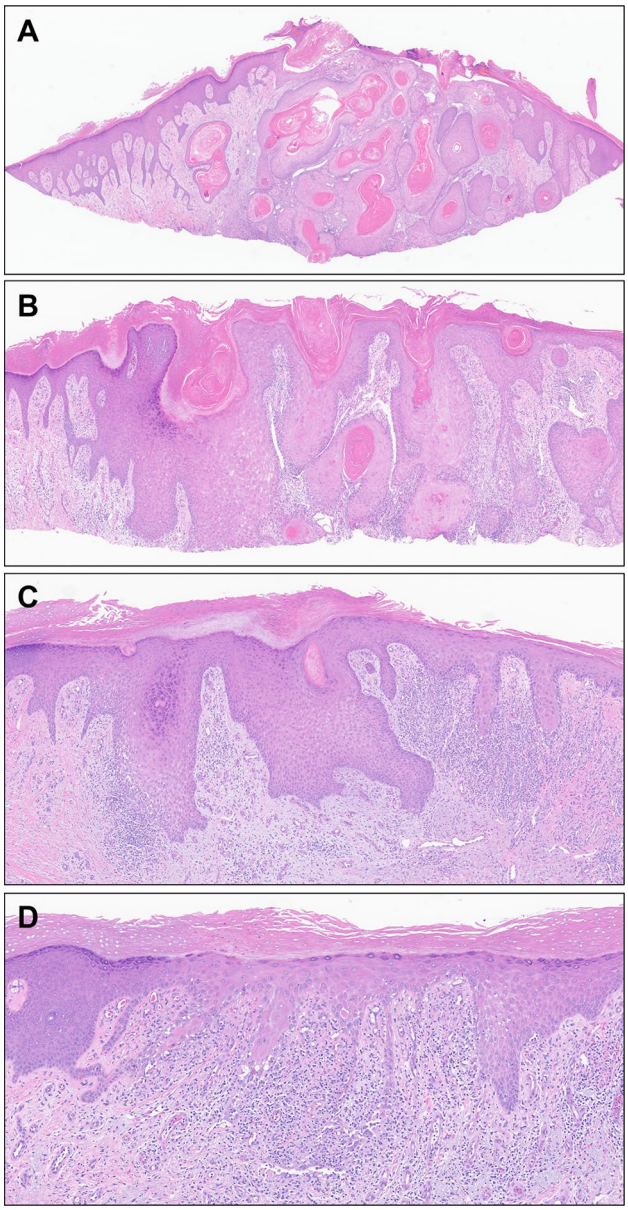


Fig. 8 Malignancy arising in association with longstanding hypertrophic lichen planus. **A** and **B** depict two invasive well-differentiated squamous cell carcinomas that developed during long-term follow-up. Other areas from the same specimens revealed changes more typical of hypertrophic lichen planus, as shown in (**C**, **D**) (**A**: H&E ×20, **B**: H&E ×40, **C**: H&E ×60, and **D**: H&E ×100)

rete ridges, resembling a saw-tooth [2, 54]. Clefting may be present ("Max-Joseph spaces"). [3, 54]. Eosinophilic globular material, corresponding to apoptotic keratinocytes, is commonly observed in the papillary dermis and basal layer of the epithelia (termed colloid bodies and Civatte bodies, respectively) [2, 14] (Fig. 1). Melanophages are another consequence of interface dermatitis [2] (Figs. 3B, 4B, and 5). The epidermis exhibits hyperorthokeratosis and irregular hyperplasia with wedge-shaped hypergranulosis [54]

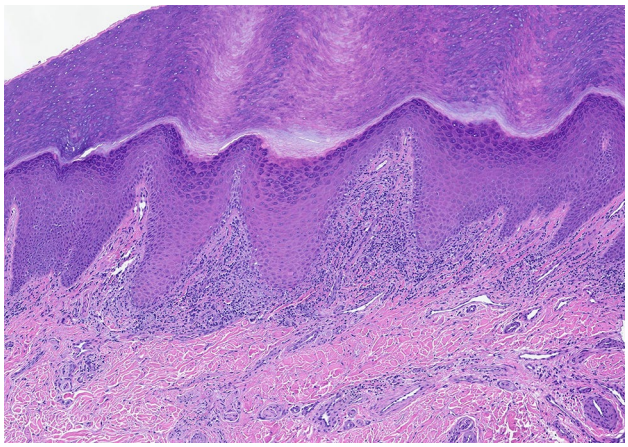


Fig. 9 Lichen planus on acral skin (H&E ×100)

(Fig. 9). Unusual histopathologic features of cutaneous LP include a predominant plasma cell infiltrate [91] and lymphocytic syringotropism, resembling syringotropic mycosis fungoides [265]. In hypertrophic LP, the inflammation is more eosinophil-rich and concentrated around the rete ridges, which tend to be more rounded than the saw-tooth rete ridges seen in classical LP [43, 54, 266]. In addition, the epidermal hyperplasia is more prominent in this variant [38] (Figs. 8 and 10).

An important differential diagnosis of LP is LDE. Clinically, the latter is more commonly polymorphous, resembling psoriasis or eczema, and tends to be devoid of WS [14]. Moreover, involvement of the mucous membranes, flexural surfaces, or nails is less likely in LDEs compared with LP [2, 4, 86], while symmetric involvement of photoexposed areas is more typical of LDE [4]. On histopathological evaluation, LDEs may be distinguished from LP by the presence of parakeratosis, eosinophils, and a higher number of apoptotic keratinocytes [13, 38, 87]. Despite these well-established differences, LP and LDE may be virtually indistinguishable in clinical practice [38]. The identification of a temporal relationship with exposure to the culprit medication is key to the diagnosis of LDEs, although this may be challenging at times.

Several other entities enter the differential diagnosis of LP (Table 2). Importantly, syphilis can mimic LP or LDEs both clinically and histopathologically. Although syphilis is associated with elongated rete ridges and plasma cells, it can deviate from this classical presentation and display a protean phenotype, underscoring the importance of maintaining a low threshold for appropriate ancillary studies [267, 268]. Lichenoid keratosis, which usually manifests with a single lesion on the trunk, is within the same histopathologic differential diagnosis as LP. The presence of seborrheic keratosis or lentigo-like changes in the epidermis adjacent to the lichenoid inflammation hints at this diagnosis in the

appropriate clinical context [2]. Furthermore, there can be clinical overlap between the papular form of atopic dermatitis and papular LP, especially in dark-complexioned patients [269, 270].

The classic DIF finding of LP is globular deposits of immunoglobulins (most commonly IgM) and complement representing cytoid bodies, as well as shaggy fibrinogen deposition at the DEJ [3, 271] (Fig. 11). The latter has been regarded as one of the best indicators for LP (especially OLP) by some authors [271–273]. The positive yield of DIF in the diagnosis of LP ranges from 37 to 97% [271]. In clinical practice, DIF has particular utility in the following scenarios: (1) differential diagnosis between erosive mucosal LP and autoimmune blistering dermatoses [2, 274], (2) differential diagnosis between LP and LE [271, 273], (3) diagnosis of LE/LP overlap syndrome, as previously discussed, and (4) diagnosis of LPPemph. In the latter, DIF typically demonstrates linear deposition of exclusively C3 or C3 and IgG along the DEJ in perilesional biopsies, while the lichenoid lesions exhibit immunofluorescence features of LP [164, 168, 275] (Fig. 6). Indirect immunofluorescence performed on 1 M NaCl-salt split skin can also support the diagnosis of LPPemph by showing autoantibodies binding to the roof of the blister [164].

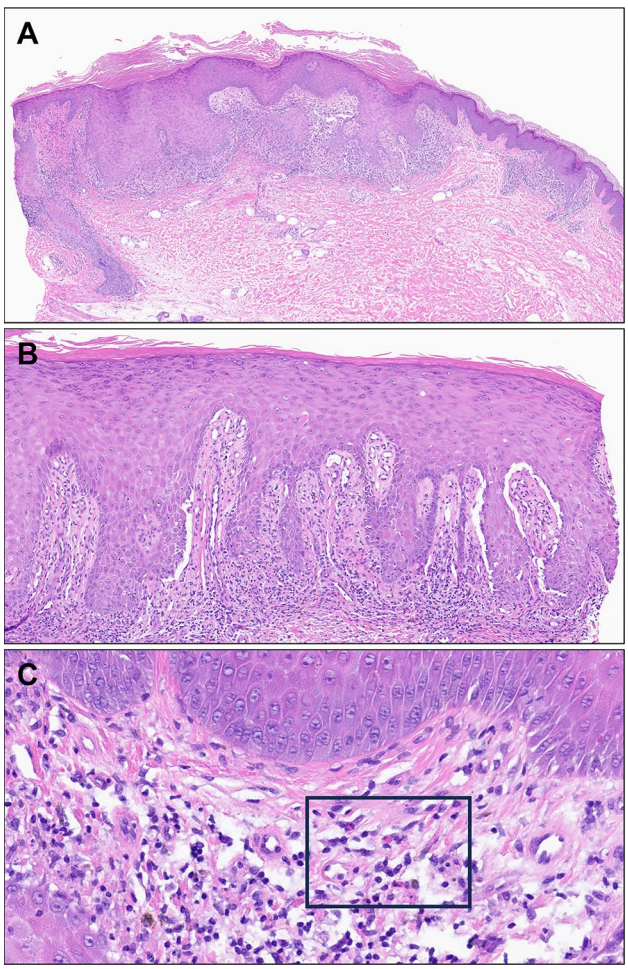


Fig. 10 Representative case of hypertrophic lichen planus, showing lichenoid inflammation with foci of subepidermal clefting (**A** and **B**). In this lichen planus variant, the inflammation is typically concentrated around the rete ridges, which assume a more rounded morphology. Furthermore, there is prominent acanthosis and admixture of eosinophils, as shown in the rectangle in (**C**) (**A**: H&E ×40, **B**: H&E ×100, and **C**: H&E ×400)

Dermoscopy is another ancillary diagnostic tool for LP and can be used to better visualize the WS, which can assume various morphologies such as reticular, rounded, globular, circular, annular, or arboriform [3, 14, 276]. Notably, the presence of WS can be interpreted as a marker of active disease as these tend to resolve following treatment [276]. Additional dermoscopic clues are vascular structures, such as red dots or lines, and diffuse, dotted, or clustered hyperpigmentation [276, 277]. Dermoscopy of LP in skin of color is characterized by the absence of vascular features and variations in WS, such as a black-hole pigmentation pattern. It has been demonstrated that dermoscopy improves the diagnosis of LP in patients with skin of color [277]. Dermoscopy may also assist in the differential diagnosis between prurigo nodularis and HLP, which can both present with pruritic nodules on the shins and ankles. Red dots and globules are more common in prurigo nodularis, whereas HLP typically exhibits gray-blue globules and comedo-like openings [278, 279].

5.2 Diagnosis and Differential Diagnosis of Mucosal Lichen Planus

Numerous diagnostic criteria have been proposed and revised for OLP; however, their interpretation is fraught with marked inter- and intraobserver variability [21, 57]. The WHO advise using clear clinical and histopathological criteria for the diagnosis of OLP [251, 280, 281]. The differential diagnosis for OLP is broad and encompasses graft-versus-host disease (GvHD), autoimmune blistering dermatoses, LE, candidiasis, and leukoplakia (particularly for plaque-like oral LP), among other conditions [38, 57] (Table 2). The distinction between OLP and oral lichenoid lesions represents a major diagnostic challenge due to their overlapping features [282]. Clinically, the presence of a culprit material, such as amalgam filling, in direct topographic relation with the lichenoid lesions appears to be the only factor distinguishing oral contact-induced lichenoid lesions from OLP, which tends to be bilateral [65, 282]. Several histopathologic findings, such as eosinophils and scattered apoptotic keratinocytes in all layers of the epithelium, have been described as being more typical of oral lichenoid lesions. However, a more recent study did not identify a significant difference in histopathological findings between OLP and oral lichenoid lesions [282]. There is some evidence that

Table 2 Nonexhaustive list of the entities within the differential diagnosis of lichen planus (LP) variants and subtypes.

LP variant/subtype	Entities in the differential diagnosis
Cutaneous LP	
Annular LP	Annular lichenoid dermatitis of youth, erythema multiforme, fungal infections, granuloma annulare
Actinic LP (LP subtropicus)	Erythema dyschromicum perstans, facial discoid dermatosis, fixed drug eruption, lupus erythematosus, melasma, syphilis
Atrophic LP	Erythema dyschromicum perstans, lichen sclerosus et atrophicus, lupus erythematosus, mycosis fungoides
Erythrodermic LP	Erythrodermic psoriasis, pityriasis rubra pilaris, mycosis fungoides
Hypopigmented LP	Hypopigmented mycosis fungoides, postinflammatory pigmentary alteration, vitiligo
Hypertrophic LP	Hypertrophic lupus erythematosus, human papilloma virus infections, lichen simplex chronicus/prurigo nodularis, pemphigoid nodularis, premalignant and malignant squamous proliferations such as squamous cell carcinoma, psoriasis
Inverse LP	Inverse psoriasis, intertrigo
Linear LP	Lichen striatus, linear atrophoderma of Moulin, linear psoriasis, inflammatory linear verrucous epidermal nevus
LP pigmentosus	Erythema dyschromicum perstans, postinflammatory pigmentary alteration
Palmoplantar LP	Palmoplantar keratoderma, palmoplantar psoriasis, tinea pedis/tinea manuum
Perforating LP	Perforating dermatoses such as reactive perforating collagenosis or perforating folliculitis
Porokeratotic LP	Disseminated superficial actinic porokeratosis, other porokeratosis variants
Vesiculobullous LP	Autoimmune bullous disorders, erythema multiforme, herpes simplex virus infections, lichenoid drug eruptions
LP overlap syndromes	
LP pemphigoides	Autoimmune bullous disorders (particularly bullous pemphigoid and paraneoplastic pemphigus), bullous LP, drug reactions, erythema multiforme
Lupus erythematosus/LP overlap	Drug reactions, hypertrophic lichen planus, other forms of cutaneous lupus erythematosus
Mucosal LP	
Oral LP	Autoimmune bullous disorders, candidiasis, erythema multiforme, graft-versus-host disease, leukoplakia, lupus erythematosus, nutritional deficiencies, oral lichenoid lesions, squamous cell carcinoma, syphilis
Genital LP	Autoimmune bullous disorders, graft-versus-host disease, intertrigo, lichen sclerosus, lichen simplex chronicus, psoriasis, squamous cell carcinoma
Esophageal LP	Autoimmune bullous disorders, candidiasis, graft-versus-host disease, reflux esophagitis, viral infections
Appendageal LP	
Lichen planopilaris	Discoid lupus erythematosus, folliculitis decalvans, other types of alopecia such as central centrifugal cicatricial alopecia, acne keloidalis nuchae, longstanding traction alopecia
Nail LP	Onychomycosis, psoriasis, onychotillomania, alopecia areata-related nail alterations, traumatic onychodystrophy, yellow nail syndrome

the mast cell count could be elevated in OLP compared with oral lichenoid lesions [65]. The diagnosis of OLP is further complicated by the fact that band-like inflammation may also be due to trauma, or encountered in lichenoid contact mucositis, LE, or infections by *Candida albicans* [122].

With regard to VVLP, the most pertinent differential diagnostic consideration is vulvar lichen sclerosus (LS), mainly given the overlap in their microscopic features [184]. Clinically, the presence of vaginal involvement strongly militates against LS [13, 179]. Other clinical features that were found useful to distinguish vulvovaginal LP from vulvar LS were pain/burning symptoms, vulvar erythema/hyperkeratosis, erosions, and presence of oral disease [283]. The differential diagnosis of genital LP also includes autoimmune bullous disorders, intertrigo, psoriasis, GvHD, and malignancies [4]. Notably, vulvar lichen simplex chronicus also needs to be considered within the differential diagnosis of VVLP. Clinically, lichen simplex chronicus generally manifests with pruritic lichenified plaques, whereas erosive VVLP tends to be associated with tender erosions accompanied by WS [284].

5.3 Diagnosis of Appendageal Lichen Planus

In LPP, histopathologic examination reveals perifollicular fibrosis and lymphocytic inflammation accompanied by cytoid bodies, typically at the level of the lower infundibulum [199] (Fig. 12). Additional findings of active disease include decrease in hair density, loss of sebaceous epithelium, "squamatization" of the basal layer, and artifactual clefting [74, 199, 201]. Established lesions show nonspecific features of end-stage scarring alopecia [201]. On

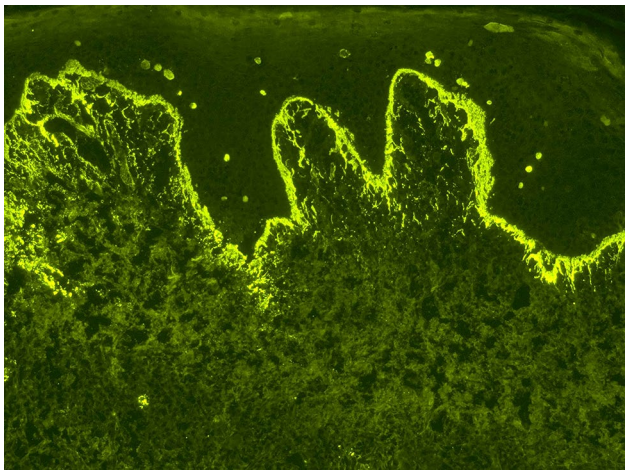


Fig. 11 Shaggy fibrinogen deposition on direct immunofluorescence examination, typical of lichen planus

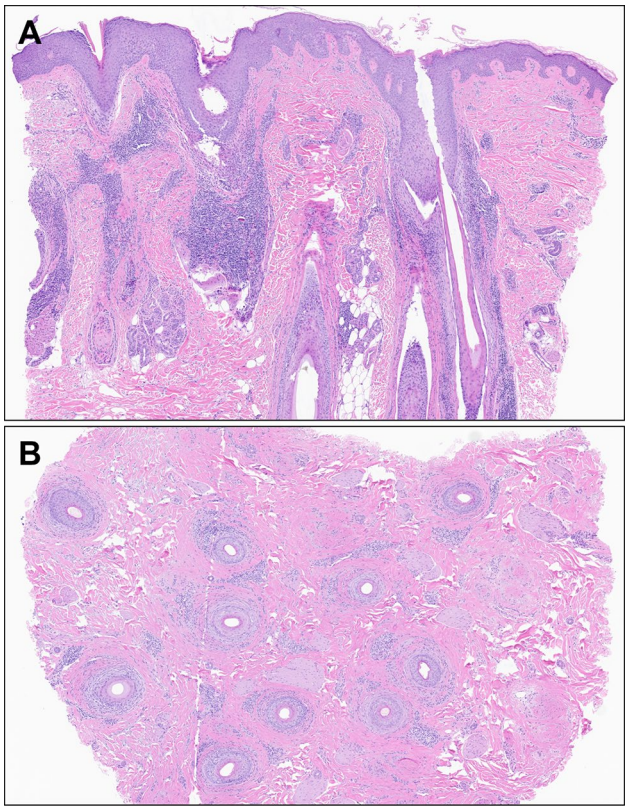


Fig. 12 Representative photomicrographs of lichen planopilaris, demonstrating perifollicular lymphocytic inflammation and concentric fibrosis on horizontal and vertical sections (**A** and **B**: H&E ×40)

trichoscopic examination, LPP demonstrates diminished follicular ostia accompanied by perifollicular erythema, scale, and keratotic plugs [74, 128].

The differential diagnosis of LPP deserves further mention (Table 2). In comparison with LPP, involvement of the trunk and eyebrows is more frequent in FFA [199]. In a histopathology study, dense perifollicular inflammatory infiltrate and concentric lamellar fibroplasia were slightly more common in LPP compared with FFA, although overall, the authors considered the histopathologic differences too subtle to permit reliable distinction between these related entities [204]. When compared with chronic cutaneous (discoid) LE, the interfollicular epidermis tends to be spared in LPP [38]. Additionally, absence of mucin, lack of basement membrane thickening, and a more superficial lymphocytic infiltrate favor a diagnosis of LPP [42, 285]. A study comparing LPP and central centrifugal cicatricial alopecia found these entities to be similar from a histopathologic and immunophenotypic standpoint, underscoring the essential role of clinicopathologic correlation in the distinction between these two forms of scarring alopecia [202]. Folliculitis decalvans is another type of cicatricial alopecia that needs to be distinguished from LPP. Clinically, the former frequently manifests with yellowish crusts, pustules, and tufted hairs, which can be highlighted using trichoscopy [286]. In contrast to LPP, folliculitis decalvans is classically regarded as a neutrophilic cicatricial alopecia. However, a mixed inflammatory infiltrate has been also described, especially in more established lesions [286]. Interestingly, a subset of patients may present with overlapping features of both entities concomitantly or consequentially, a condition known as "folliculitis decalvans lichen planopilaris phenotypic spectrum" [287, 288].

The diagnosis of nail LP relies on histopathologic confirmation, especially when nail involvement is isolated or there is uncertainty about the diagnosis [39, 69]. Lateral longitudinal biopsies have the advantage of illustrating alterations of both the nail matrix and nail bed [39, 69]. Microscopic findings include diffuse increase in granular layer, sawtooth-shaped hyperplasia, lichenoid inflammation, and disappearance of the keratogenous zone [39, 69, 214]. Nail plate fraying, corresponding to detachment of individual onychocytes from the nail plate, is a recently reported feature [214]. In nail LP, parakeratosis and spongiosis are more common than in LP of the skin [39]. Onychoscopy has the potential to assist in early diagnosis [289]. Onychomycosis can present with clinical findings similar to nail LP, underscoring the need for fungal testing when working up onychodystrophy [4]. Other entities within the differential diagnosis are psoriasis, onychotillomania, alopecia areata-related nail alterations, and trauma [69, 290] (Table 2). When confronted with a differential diagnosis between nail involvement by psoriasis and LP, pterygium strongly favors the latter [217].

6 Treatment

Management of LP requires a stepwise approach, starting with topical agents whenever feasible and proceeding to phototherapy and systemic therapies as indicated [291]. Many LP variants are notorious for a chronic relapsing course and treatment refractoriness, necessitating multiple courses of treatment with different modalities [16, 83, 292]. For many of the treatment options, there is a dearth of evidence [293]. Systemic agents, in particular, can have undesirable effects; thus, benefit-to-risk ratios need to be weighed carefully before commencing treatment [16, 294]. It is important to note that the treatment response may differ on the basis of the specific clinical variant, which in turn has implications regarding treatment selection. In particular, caution needs to be exercised in long-term treatment of HLP using topical calcineurin inhibitors, which have been associated with a theoretical risk of malignancy.

6.1 Treatment of Cutaneous Lichen Planus

Cutaneous LP frequently undergoes spontaneous remission, which hampers the objective evaluation of treatment efficacy [295]. The goal of therapy is to hasten the lesion resolution and decrease pruritus [295]. In a literature review, topical corticosteroids and topical calcineurin inhibitors (TCIs) had a high quality of evidence [296]. In a recent study on patients with cutaneous LP involving more than 10% body surface area, high-dose clobetasol propionate resulted in complete remission in almost two-thirds of patients within 6 weeks following commencement of treatment [134]. There are several reports of treatment of LP with retinoids, such as isotretinoin, alitretinoin, and acitretin [16, 297–300]. In a double-blind study, acitretin resulted in higher rates of remission compared with placebo [301]. In a review on LP therapeutics [16], two other agents were assigned a level of evidence of IB based on double-blind controlled trials, namely sulfasalazine [302] and griseofulvin [303]. Numerous other medications, such as azathioprine [304], metronidazole [305], itraconazole [306], methotrexate [307–309], mycophenolate mofetil [310], cyclosporine [23, 311, 312], leflunomide [56], colchicine [313], cyclophosphamide [314], thalidomide [157], enoxaparin [315], and intralesional injection of Bacillus Calmette–Guérin polysaccharide nucleic acid combined with vitamin D [316], have been utilized in cutaneous LP with variable results, and are regarded as third-line therapies [13]. Of these medications, methotrexate is noteworthy since there are multiple studies documenting its effectiveness in various forms of LP. The dose of oral methotrexate most commonly used for LP in the literature is 15 mg per week; however, the preferred dosage ranged from 2.5 mg to 20 mg per week [307–309, 317–322]. Dupilumab may be effective based on case reports [291, 323]. Although TNF-α inhibitors are generally known as triggers for LDEs, there are rare reports of LP improvement with these agents [174, 324].

Narrow-band ultraviolet B (NB-UVB) has an established track record in the treatment of LP and is thought to act by inhibiting antigen presentation, inducing apoptosis of T cells, and triggering an increase of immunosuppressive cytokines [23, 294, 325, 326]. In a randomized clinical trial on generalized LP, NB-UVB was more effective compared with systemic steroids [327]. In another study, approximately three-quarters of patients with cutaneous LP had at least 90% subsidence of lesions and most of the patients remained free of disease for at least 4.8 years after one treatment course, indicating the prolonged therapeutic impact of NB-UVB [328]. In light of these data, some authors recommend that NB-UVB be considered before systemic medications [23, 326]. The efficacy is comparable for psoralen plus ultraviolet A and NB-UVB; however, the latter is preferred due to a more favorable side-effect profile [326]. Other treatment modalities described in LP are transcutaneous electrical nerve stimulation [329] and fractional laser [330].

Management of immune checkpoint inhibitor-related lichenoid eruptions deserves further mention. Mild and moderate presentations are typically treated with high-potency topical corticosteroids. For severe forms, however, systemic corticosteroids and cessation of the causative immune checkpoint inhibitor(s) may be indicated [331, 332].

6.2 Treatment of Mucosal Lichen Planus

Treatment of mucosal LP is challenging, often associated with transient improvement and short-lived remissions [13, 174, 333]. Therefore, treatment is mainly palliative and aimed at minimizing pain and discomfort [334]. For OLP, topical superpotent corticosteroids are the treatment of choice and considered generally safe, with rare side effects such as candidiasis and mucosal atrophy [3, 12, 57, 175, 335–338]. Systemic corticosteroids are used to obtain rapid symptom resolution in extensive and/or erosive OLP [57, 337]. In OLP cases that are unresponsive to topical steroids, TCIs may be effective [3, 339]. In a review article, TCIs were assigned a level of evidence of IA in the context of OLP [16]. It has been suggested that TCIs be considered before initiation of systemic corticosteroids [340]. In studies on OLP, TCIs were well tolerated, with only mild and transient adverse effects such as irritation or burning [336, 339, 341]. A notable treatment option described in the literature for steroid-refractory oral LP cases is cyclosporine administered in a "swish and spit" technique [342, 343]. Topical retinoids may also be an alternative to corticosteroids [16, 344].

Several second/third-line medications, including hydroxychloroquine [345–347], oral tacrolimus [348], sirolimus [349], mycophenolate mofetil [350], methotrexate [322], thalidomide [351, 352], irsogladine maleate [353], etanercept [354], intravenous immunoglobulin [355], and metronidazole [356], have been found to be variably effective in OLP. Of these agents, hydroxychloroquine has a more established track record as a safe and effective treatment [345–347] and is regarded by some experts as one of the first-line systemic treatments in OLP [57]. Other modalities that play a potential role in OLP therapeutics include plasma rich in growth factors [357], photodynamic therapy [358, 359], extracorporeal photochemotherapy [360, 361], and lasers [362, 363]. Management recommendations also include dental care, lidocaine mouthwash, avoidance of smoking and irritant factors, and replacement of amalgam restorations, as indicated [3, 4, 361].

Treatment of genital LP is fundamentally similar to treatment of OLP. Daily high-potency topical corticosteroids followed by a less frequent maintenance regimen constitute the first line of treatment [3, 16]. For vaginal involvement, corticosteroid pessaries can be used [173, 181]. Multiple studies documented efficacy of TCIs [173, 364], although their benefit does not appear to be sustained after cessation of treatment [180]. Management of VVLP unresponsive to the initial topical treatments is very difficult [182, 341]. In a study of severe VVLP, systemic immunosuppression was necessary in nearly half of the patients [185]. The most commonly used systemic medications are corticosteroids, mycophenolate mofetil, and methotrexate, but the literature lacks high-quality evidence to favor one treatment over the other in severe genital LP [173, 317, 321, 365, 366]. Other modalities that have been attempted include acitretin [367], hydroxychloroquine [368], rituximab [369], and photodynamic therapy [370]. In severe genital LP, surgical lysis of adhesions or dilation may be indicated [173, 180]. For esophageal LP, swallowed corticosteroids represent an option, but endoscopic dilation may be needed to address the strictures [191, 192].

6.3 Treatment of Appendageal Lichen Planus

The primary objectives in the treatment of LPP are promptly stopping the inflammatory process, alleviating symptoms, and halting the progression of hair loss [128]. For limited scalp involvement, intralesional and topical corticosteroids are the mainstay of treatment [16, 128]. A therapeutic ladder, wherein systemic agents are added in cases that are treatment-refractory or have extensive involvement, has been proposed [128, 199]. A histopathologic study found that the "normal-appearing" scalp areas are also affected on a microscopic level, lending support to systemic therapy [371]. The recommended first-line systemic agents vary between publications, with most authors recommending hydroxychloroquine [74, 128, 285, 372]. In a case series of 29 patients, the only treatment that yielded statistically significant outcomes in LPP was tetracycline [201]. In a systematic review, cyclosporine was the most efficacious medication, followed by systemic corticosteroids [373]. Other modalities employed in the treatment of LPP with limited evidence include low-dose naltrexone [374, 375], oral minoxidil [376], pioglitazone hydrochloride [377], mycophenolate mofetil [197], retinoids [378], methotrexate [379], thalidomide [380], NB-UVB fiber-optic brush [381], laser [382], and hair transplant (ensuing cessation of disease activity) [128].

Nail LP can follow an unpredictable and rapidly progressive course, leading to a substantial functional impact and permanent nail dystrophy in a subset of cases [215, 383]. In fact, nail LP has been considered a "true nail emergency," prompting early initiation of treatment to prevent scarring [69, 384]. Nevertheless, evidence-based therapeutic options for this condition are limited [16]. Several experts advocate the use of triamcinolone acetonide as the first-line treatment, with the mode of administration (intralesional every 4–5 weeks versus intramuscular) depending on the extent of involvement ($\leq$ 3 nails versus $>$ 3 nails, respectively) [39, 215, 385]. Intralesional triamcinolone provides targeted delivery of the anti-inflammatory effect to the nail matrix while avoiding systemic adverse effects, however, may be painful [264]. Nail bed involvement and patient reluctance are scenarios where intramuscular route is applicable [39, 69, 215]. Oral retinoids such as alitretinoin or acitretin represent second-line treatment options for LP affecting more than three nails [1, 218, 385]. In persistent cases, consideration may be given to corticosteroid-sparing immunosuppressive agents, such as cyclosporine, or TNF-α inhibitors [215, 385–387]. Localized bath-PUVA has also been tried [388]. In general, topical treatments lack efficacy due to limited penetration to the nail [39]. However, TCIs were successfully employed in a case series of nail LP [383].

6.4 Treatment of Lichen Planus with JAK Inhibitors

There is a robust biologic rationale for using inhibitors of the JAK–STAT pathway in LP, as previously elaborated on in Sect. 2.2.2.1. Accordingly, therapeutic targeting of this pathway has been used with success in virtually all LP forms [389], such as cutaneous [390, 391] (including HLP [392, 393]), oral [176, 394–398], genital [398, 399], esophageal [191, 397, 400], nail LP [216, 292, 401], and LPP/FFA [75, 402–404]. Based on a recent systematic review, partial or complete resolution was obtained in 73.3% of the instances of LP treated with JAK inhibitors [81]. Several advantages of these medications stand out from review of the literature. First, many series described rapid alleviation of symptoms,

within days to weeks [216, 390, 391, 394]. Second, JAK inhibitors demonstrated efficacy in numerous cases of LP that were refractory to other therapies [75, 390, 391, 398, 403, 404]. Third, JAK inhibitors were generally well tolerated in LP, without development of serious side effects [75, 216, 390, 391, 400]. However, it is important to be cognizant of the warnings about the increased risk of certain adverse effects associated with oral JAK inhibitors, such as venous thromboembolism, major cardiovascular events, malignancies, and serious infections [399, 405]. Finally, many JAK inhibitors are also available in topical form, allowing targeted treatment of the involved body site [391, 402, 404]. Indeed, topical JAK inhibitors have been successfully used in the treatment of both cutaneous LP [391] and LPP/FFA [402, 404]. Importantly, different medications have variable activity against different members of the JAK family, and it remains to be determined which JAK-related pathway needs to be targeted for optimal results in LP [191, 396].

6.5 Treatment of Lichen Planus with Phosphodiesterase-4 Inhibitors

Apremilast is a small-molecule PDE-4 inhibitor that has been employed in the management of psoriasis and Behçet disease as an immunomodulator [177, 406, 407]. It acts by increasing the intracellular levels of cyclic AMP and consequently inhibiting the transcription of pro-inflammatory cytokines [296, 406, 408]. Therapeutic utility of PDE-4 inhibitors was mostly reported in the context of OLP, including recalcitrant cases [57, 334, 409, 410]. Crushed apremilast compounded into solution for topical application or swish-and-spit mouthwash may be an alternative to systemic administration [407]. Apremilast also holds potential in other LP forms [408, 411, 412]. In an open-label study, ten patients with moderate to severe LP were treated using oral apremilast for 12 weeks. With regard to the secondary endpoints, which included response of mucosal disease and impact on QoL among other parameters, all patients showed therapeutic improvement [406]. In a multicenter study on 11 patients with OLP who had a history of failure with other therapies, a symptomatic and clinical improvement with apremilast treatment was noted in 6 patients (55%) at the 12-week mark [409]. Overall, the safety profile and tolerability of apremilast are well tolerated [177, 334, 406], with headache and nausea being the most frequent adverse effects described in the studies on LP [406, 411].

6.6 Treatment of Lichen Planus with Interleukin-23/Interleukin-17 Pathway Inhibitors

The IL-23/IL-17 pathway is central to pathogenesis of LP based on research indicating enhanced Th17 cytokine expression, lesional infiltration of IL-17$^+$ and IL-23$^+$ mast cells, and increased levels of circulating IL-17 and IL-23 in this condition [66, 116, 413–417]. Accordingly, targeting of the IL-23/IL-17 axis was shown to result in improvement of cutaneous LP and/or OLP [116, 175, 418], genital LP [185, 187], LPPemph [168, 275], and LPP [419]. In a cohort of 24 women with severe VVLP treated with tildrakizumab, an IL-23 antagonist, 21 patients (87.5%) achieved sustained remission during the study follow-up duration. Of note, remission had not been achieved in 17 patients until tildrakizumab was introduced. The most common adverse effect was vulvovaginal candidiasis followed by injection-site reactions, encountered in 20.8% and 8.3% of the patients, respectively [185].

7 Quality of Life in Lichen Planus

Quality of life (QoL) impairment has been documented in all forms of LP [18, 370]. Oral LP has a major impact on eating, drinking, and procedures of oral hygiene, particularly in its erosive and erythematous forms [338, 359]. Genital LP may be life-disruptive by causing pruritus, burning, dyspareunia, sexual and genitourinary dysfunction, and profound psychological effects [145, 180, 367, 370, 408]. In a single-center study of vulvar LP, the QoL impairment failed to improve despite treatment, indicating that the current treatments are insufficient [184]. There appears to be a bidirectional link between psychosocial stress and LP [13]. Stressful life events and emotional stress related to family members were shown to trigger symptoms [18, 88, 420]. Depression and anxiety are more common in patients with LP [14, 18, 421]. Impaired QoL also occurs in LPP, which has been linked to a high prevalence of sleep disturbances, depression, and anxiety [74, 128]. Despite the established impact of LP on QoL, there remains a paucity of widely utilized and uniform, patient-reported outcome measures and lesional or multisite scoring systems [81, 422, 423].

8 Conclusion

Lichen planus is an idiopathic inflammatory disease with a heterogeneous clinical presentation, affecting approximately 0.5–1% of the population. There is mounting evidence that LP is not an isolated skin disorder, given the reports associating LP with autoimmune conditions, glucose intolerance, dyslipidemia, and cardiovascular disorders. Cutaneous hypertrophic and mucosal forms of LP are at a heightened risk for malignant transformation. Familiarity with these potential associations in conjunction with long-term follow-up and regular screening could lead to a timely diagnosis and management of concomitant conditions. The multisite

involvement of LP requires a multidisciplinary approach. The frequent QoL impairment in LP underscores the need for a comprehensive approach including psychological evaluation and support. More recently, advent of innovative and pathogenesis-driven treatment modalities has expanded therapeutic options for patients with LP.

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References

1. Didona D, Pumnea T, von Braunmühl T, Wolf R. Effective use of alitretinoin in acitretin-refractory nail lichen planus. Int J Dermatol. 2022;61(5):e174–6.
2. Deschaine MA, Lehman JS. The interface reaction pattern in the skin: an integrated review of clinical and pathological features. Hum Pathol. 2019;91:86–113.
3. Ioannides D, Vakirlis E, Kemeny L, Marinovic B, Massone C, Murphy R, et al. European S1 guidelines on the management of lichen planus: a cooperation of the European Dermatology Forum with the European Academy of Dermatology and Venereology. J Eur Acad Dermatol Venereol. 2020;34(7):1403–14.
4. Boch K, Langan EA, Kridin K, Zillikens D, Ludwig RJ, Bieber K. Lichen Planus. Front Med (Lausanne). 2021;8: 737813.
5. Nguyen B, Perez AG, Elgart GW, Elman SA. Lichen planus after COVID-19 infection and vaccination: a systematic review. J Eur Acad Dermatol Venereol. 2023;37(3):e278–81.
6. Zou H, Daveluy S. Lichen planus after COVID-19 infection and vaccination. Arch Dermatol Res. 2023;315(2):139–46.
7. Al-Khenaizan S. Lichen planus occurring after hepatitis B vaccination: a new case. J Am Acad Dermatol. 2001;45(4):614–5.
8. An I, Demir V, İbiloğlu İ, Akdeniz S. Lichen planus secondary to rabies vaccination. Dermatol Online J. 2017;23(3):1–3.
9. Caccavale S, Caccavale T, La Montagna M. Lichen planus after rabies vaccination: an example of isovaccinetopic response. Int J Dermatol. 2017;56(2):e40–1.
10. Dangien A, Darbord D, Chanal J, Wendling J, Pantoja C, Oules B, et al. SARS-CoV-2 vaccination may trigger and exacerbate mucosal lichen planus. J Eur Acad Dermatol Venereol. 2023;37(9):e1094–6.
11. Rosengard HC, Wheat CM, Tilson MP, Cuda JD. Lichen planus following tetanus-diphtheria-acellular pertussis vaccination: a case report and review of the literature. SAGE Open Med Case Rep. 2018;6:2050313x17750335.
12. Thongprasom K, Luengvisut P, Wongwatanakij A, Boonjatturus C. Clinical evaluation in treatment of oral lichen planus with topical fluocinolone acetonide: a 2-year follow-up. J Oral Pathol Med. 2003;32(6):315–22.
13. Lehman JS, Tollefson MM, Gibson LE. Lichen planus. Int J Dermatol. 2009;48(7):682–94.
14. Tziotzios C, Lee JYW, Brier T, Saito R, Hsu CK, Bhargava K, et al. Lichen planus and lichenoid dermatoses: clinical overview and molecular basis. J Am Acad Dermatol. 2018;79(5):789–804.
15. Rennie CE, Dwivedi RC, Khan AS, Agrawal N, Ziyada W. Lichen planus of the larynx. J Laryngol Otol. 2011;125(4):432–5.
16. Tziotzios C, Brier T, Lee JYW, Saito R, Hsu CK, Bhargava K, et al. Lichen planus and lichenoid dermatoses: Conventional and emerging therapeutic strategies. J Am Acad Dermatol. 2018;79(5):807–18.
17. Pannell RS, Fleming DM, Cross KW. The incidence of molluscum contagiosum, scabies and lichen planus. Epidemiol Infect. 2005;133(6):985–91.
18. Fiocco Z, Kupf S, Patzak L, Kämmerer T, Pumnea T, French LE, et al. Quality of life and psychopathology in lichen planus: a neglected disease burden. Acta Derm Venereol. 2021;101(12):adv00619.
19. Brewer JD, Ekdawi NS, Torgerson RR, Camilleri MJ, Bruce AJ, Rogers RS 3rd, et al. Lichen planus and cicatricial conjunctivitis: disease course and response to therapy of 11 patients. J Eur Acad Dermatol Venereol. 2011;25(1):100–4.
20. Joshi TP, Zhu H, Naqvi Z, Holla S, Duruewuru A, Ren V. Prevalence of lichen planopilaris in the United States: a cross-sectional study of the All of Us research program. JAAD Int. 2022;8:69–70.
21. González-Moles M, Warnakulasuriya S, González-Ruiz I, González-Ruiz L, Ayén Á, Lenouvel D, et al. Worldwide prevalence of oral lichen planus: a systematic review and meta-analysis. Oral Dis. 2021;27(4):813–28.
22. Halonen P, Jakobsson M, Heikinheimo O, Gissler M, Pukkala E. Incidence of lichen planus and subsequent mortality in Finnish Women. Acta Derm Venereol. 2020;100(17):adv00303.
23. Schwager Z, Stern M, Cohen J, Femia A. Clinical epidemiology and treatment of lichen planus: A retrospective review of 2 tertiary care centers. J Am Acad Dermatol. 2019;81(6):1397–9.
24. Li C, Tang X, Zheng X, Ge S, Wen H, Lin X, et al. Global prevalence and incidence estimates of oral lichen planus: a systematic review and meta-analysis. JAMA Dermatol. 2020;156(2):172–81.
25. Zendell K. Genital lichen planus: update on diagnosis and treatment. Semin Cutan Med Surg. 2015;34(4):182–6.
26. Pathoulas JT, Flanagan KE, Walker CJ, Collins MS, Ali S, Pupo Wiss IM, et al. A multicenter descriptive analysis of 270 men with frontal fibrosing alopecia and lichen planopilaris in the United States. J Am Acad Dermatol. 2023;88(4):937–9.
27. Trager MH, Lavian J, Lee EY, Gary D, Jenkins F, Christiano AM, et al. Prevalence estimates for lichen planopilaris and frontal fibrosing alopecia in a New York City health care system. J Am Acad Dermatol. 2021;84(4):1166–9.

28. Lobato-Berezo A, Iglesias-Sancho M, Rodríguez-Lomba E, Mir-Bonafé JF, Velasco-Tamariz V, Porriño-Bustamante ML, et al. Frontal fibrosing alopecia in men: a multicenter study of 39 patients. J Am Acad Dermatol. 2022;86(2):481–4.
29. Prasad S, Marks DH, Burns LJ, De Souza B, Flynn EA, Scheinman P, et al. Patch testing and contact allergen avoidance in patients with lichen planopilaris and/or frontal fibrosing alopecia: a cohort study. J Am Acad Dermatol. 2020;83(2):659–61.
30. Dicle O, Celik-Ozenci C, Sahin P, Pfannes EKB, Vogt A, Altinok BN, et al. Differential expression of mTOR signaling pathway proteins in lichen planopilaris and frontal fibrosing alopecia. Acta Histochem. 2018;120(8):837–45.
31. Del Duca E, Ruano Ruiz J, Pavel AB, Sanyal RD, Song T, Gay-Mimbrera J, et al. Frontal fibrosing alopecia shows robust T helper 1 and Janus kinase 3 skewing. Br J Dermatol. 2020;183(6):1083–93.
32. Doche I, Wilcox GL, Ericson M, Valente NS, Romiti R, McAdams BD, et al. Evidence for neurogenic inflammation in lichen planopilaris and frontal fibrosing alopecia pathogenic mechanism. Exp Dermatol. 2020;29(3):282–5.
33. Merhy R, Sarkis AS, Assaf J, Afiouni R, Zeinaty P, Kechichian E, et al. Pediatric lichen planus: a systematic review of 985 published cases. Int J Dermatol. 2022;61(4):416–21.
34. Bevans SL, Theos AJ, Fowler PG, Pavlidakey PG, Stoll M, Sami N. Pediatric ocular lichen planus and lichen planopilaris: One new case and a review of the literature. Pediatr Dermatol. 2018;35(6):859–63.
35. Pandhi D, Singal A, Bhattacharya SN. Lichen planus in childhood: a series of 316 patients. Pediatr Dermatol. 2014;31(1):59–67.
36. Balasubramaniam P, Ogboli M, Moss C. Lichen planus in children: review of 26 cases. Clin Exp Dermatol. 2008;33(4):457–9.
37. Mohandesi NA, Berry NA, Tollefson MM, Lehman JS, Davis DMR. Pediatric lichen planus: a single-center retrospective review of 26 patients with follow up. Pediatr Dermatol. 2023;40(1):84–9.
38. Gorouhi F, Davari P, Fazel N. Cutaneous and mucosal lichen planus: a comprehensive review of clinical subtypes, risk factors, diagnosis, and prognosis. ScientificWorldJournal. 2014;2014: 742826.
39. Grover C, Kharghoria G, Baran R. Nail lichen planus: A review of clinical presentation, diagnosis and therapy. Ann Dermatol Venereol. 2022;149(3):150–64.
40. Belhareth K, Korbi M, Belhadjali H, Soua Y, Moussa A, Youssef M, et al. Pediatric lichen planus pigmentosus possibly triggered by mercury dental amalgams. Pediatr Dermatol. 2020;37(2):375–6.
41. Ghosh A, Bhunia D, Rudra O, Agarwal M. Perforating lichen planus in an adolescent boy: a rare phenomenon. Indian J Dermatol. 2017;62(1):100–1.
42. Christensen KN, Lehman JS, Tollefson MM. Pediatric lichen planopilaris: clinicopathologic study of four new cases and a review of the literature. Pediatr Dermatol. 2015;32(5):621–7.
43. Idriss MH, Barbosa N, Chang MB, Gibson L, Baum CL, Vidal NY. Concomitant hypertrophic lichen planus and squamous cell carcinoma: clinical features and treatment outcomes. Int J Dermatol. 2022;61(12):1527–31.
44. El-Rifaie AA, Rashed LA, Doss RW, Osman ST. MicroRNAs in cutaneous lichen planus. Clin Exp Dermatol. 2017;42(8):898–901.
45. Kofoed ML, Wantzin GL. Familial lichen planus. More frequent than previously suggested? J Am Acad Dermatol. 1985;13(1):50–4.
46. Sandhu K, Handa S, Kanwar AJ. Familial lichen planus. Pediatr Dermatol. 2003;20(2):186.
47. Gibstine CF, Esterly NB. Lichen planus in monozygotic twins. Arch Dermatol. 1984;120(5):580.
48. Setterfield JF, Neill S, Shirlaw PJ, Theron J, Vaughan R, Escudier M, et al. The vulvovaginal gingival syndrome: a severe subgroup of lichen planus with characteristic clinical features and a novel association with the class II HLA DQB1*0201 allele. J Am Acad Dermatol. 2006;55(1):98–113.
49. Carrozzo M, Francia Celle P, Gandolfo S, Carbone M, Conrotto D, Fasano ME, et al. Increased frequency of HLA-DR6 allele in Italian patients with hepatitis C virus-associated oral lichen planus. Br J Dermatol. 2001;144(4):803–8.
50. Pavlovsky L, Israeli M, Sagy E, Berg AL, David M, Shemer A, et al. Lichen planopilaris is associated with HLA DRB1*11 and DQB1*03 alleles. Acta Derm Venereol. 2015;95(2):177–80.
51. Liu J, Ye Z, Mayer JG, Hoch BA, Green C, Rolak L, et al. Phenome-wide association study maps new diseases to the human major histocompatibility complex region. J Med Genet. 2016;53(10):681–9.
52. Liu W, Li M, Zhang X, Zhou Z, Shen Z, Shen X. Association of polymorphisms in Th1/Th2-related cytokines (IFN-γ, TGFβ1, IL-1β, IL-2, IL-4, IL-18) with oral lichen planus: A pooled analysis of case-control studies. J Dent Sci. 2023;18(2):560–6.
53. Nagao Y, Nishida N, Toyo-Oka L, Kawaguchi A, Amoroso A, Carrozzo M, et al. Genome-wide association study identifies risk variants for lichen planus in patients with hepatitis C virus infection. Clin Gastroenterol Hepatol. 2017;15(6):937-44.e5.
54. Whittington CP, Saleh JS, Bresler SC, Patel RM. Hypertrophic lichen planus: an up-to-date review and differential diagnosis. Arch Pathol Lab Med. 2024;148(6):659–65.
55. Sugerman PB, Satterwhite K, Bigby M. Autocytotoxic T-cell clones in lichen planus. Br J Dermatol. 2000;142(3):449–56.
56. Sardana K, Gupta A. A recalcitrant case of ulcerative lichen planus responsive to leflunomide. Clin Exp Dermatol. 2020;45(3):340–1.
57. Louisy A, Humbert E, Samimi M. Oral lichen planus: an update on diagnosis and management. Am J Clin Dermatol. 2024;25(1):35–53.
58. Norris DA. Cytokine modulation of adhesion molecules in the regulation of immunologic cytotoxicity of epidermal targets. J Invest Dermatol. 1990;95(6 Suppl):111s-s120.
59. Wenzel J, Tüting T. An IFN-associated cytotoxic cellular immune response against viral, self-, or tumor antigens is a common pathogenetic feature in "interface dermatitis." J Invest Dermatol. 2008;128(10):2392–402.
60. Weber B, Schlapbach C, Stuck M, Simon HU, Borradori L, Beltraminelli H, et al. Distinct interferon-gamma and interleukin-9 expression in cutaneous and oral lichen planus. J Eur Acad Dermatol Venereol. 2017;31(5):880–6.
61. Shen Z, Du G, Zhou Z, Liu W, Shi L, Xu H. Aberrant expression of interleukin-22 and its targeting microRNAs in oral lichen planus: a preliminary study. J Oral Pathol Med. 2016;45(7):523–7.
62. Georgescu SR, Mitran CI, Mitran MI, Nicolae I, Matei C, Ene CD, et al. Oxidative stress in cutaneous lichen planus-a narrative review. J Clin Med. 2021;10(12):1–15.
63. Paulusová V, Laco J, Drízhal I, Slezák R. Expression of matrix metalloproteinase 9 in patients with oral lichen planus. Acta Medica (Hradec Kralove). 2012;55(1):23–6.
64. Domingues R, de Carvalho GC, Aoki V, da Silva Duarte AJ, Sato MN. Activation of myeloid dendritic cells, effector cells and regulatory T cells in lichen planus. J Transl Med. 2016;14(1):171.
65. Ramalingam S, Malathi N, Thamizhchelvan H, Sangeetha N, Rajan ST. Role of mast cells in oral lichen planus and oral lichenoid reactions. Autoimmune Dis. 2018;2018:7936564.
66. Hobo A, Harada K, Maeda T, Uchiyama M, Irisawa R, Yamazaki M, et al. IL-17-positive mast cell infiltration in the lesional skin

of lichen planopilaris: Possible role of mast cells in inducing inflammation and dermal fibrosis in cicatricial alopecia. Exp Dermatol. 2020;29(3):273–7.
67. Mizukawa Y, Yamazaki Y, Shiohara T. Leakage of sweat into the dermo-epidermal junction as a possible trigger for lichen planus lesion development. Arch Dermatol Res. 2019;311(1):71–82.
68. Frew JW. We need to talk about Notch: Notch dysregulation as an epiphenomenon in inflammatory skin disease. Br J Dermatol. 2019;180(2):431–2.
69. Gupta MK, Lipner SR. Review of nail lichen planus: epidemiology, pathogenesis, diagnosis, and treatment. Dermatol Clin. 2021;39(2):221–30.
70. Domingues R, Pietrobon AJ, Carvalho GC, Pereira NZ, Pereira NV, Sotto MN, et al. Lichen planus: altered AIM2 and NLRP1 expression in skin lesions and defective activation in peripheral blood mononuclear cells. Clin Exp Dermatol. 2019;44(4):e89–95.
71. de Carvalho GC, Hirata FYA, Domingues R, Figueiredo CA, Zaniboni MC, Pereira NV, et al. Up-regulation of HMGB1 and TLR4 in skin lesions of lichen planus. Arch Dermatol Res. 2018;310(6):523–8.
72. Hussein MR. Evaluation of angiogenesis in normal and lichen planus skin by CD34 protein immunohistochemistry: preliminary findings. Cell Biol Int. 2007;31(10):1292–5.
73. Siponen M, Bitu CC, Al-Samadi A, Nieminen P, Salo T. Cathepsin K expression is increased in oral lichen planus. J Oral Pathol Med. 2016;45(10):758–65.
74. Larkin SC, Cantwell HM, Imhof RL, Torgerson RR, Tolkachjov SN. Lichen planopilaris in women: a retrospective review of 232 women seen at mayo clinic from 1992 to 2016. Mayo Clin Proc. 2020;95(8):1684–95.
75. Yang CC, Khanna T, Sallee B, Christiano AM, Bordone LA. Tofacitinib for the treatment of lichen planopilaris: a case series. Dermatol Ther. 2018;31(6): e12656.
76. Karnik P, Tekeste Z, McCormick TS, Gilliam AC, Price VH, Cooper KD, et al. Hair follicle stem cell-specific PPAR-gamma deletion causes scarring alopecia. J Invest Dermatol. 2009;129(5):1243–57.
77. Imanishi H, Ansell DM, Chéret J, Harries M, Bertolini M, Sepp N, et al. Epithelial-to-mesenchymal stem cell transition in a human organ: lessons from lichen planopilaris. J Invest Dermatol. 2018;138(3):511–9.
78. Welz-Kubiak K, Kobuszewska A, Reich A. IL-31 is overexpressed in lichen planus but its level does not correlate with pruritus severity. J Immunol Res. 2015;2015: 854747.
79. Shao S, Tsoi LC, Sarkar MK, Xing X, Xue K, Uppala R, et al. IFN-γ enhances cell-mediated cytotoxicity against keratinocytes via JAK2/STAT1 in lichen planus. Sci Transl Med. 2019;11(511):1–26.
80. Pietschke K, Holstein J, Meier K, Schäfer I, Müller-Hermelink E, Gonzalez-Menendez I, et al. The inflammation in cutaneous lichen planus is dominated by IFN-ϒ and IL-21-A basis for therapeutic JAK1 inhibition. Exp Dermatol. 2021;30(2):262–70.
81. Abduelmula A, Bagit A, Mufti A, Yeung KCY, Yeung J. The use of janus kinase inhibitors for lichen planus: an evidence-based review. J Cutan Med Surg. 2023;27(3):271–6.
82. Alves de Medeiros AK, Speeckaert R, Desmet E, Van Gele M, De Schepper S, Lambert J. JAK3 as an emerging target for topical treatment of inflammatory skin diseases. PLoS ONE. 2016;11(10): e0164080.
83. Boyd AS, Neldner KH. Lichen planus. J Am Acad Dermatol. 1991;25(4):593–619.
84. Beer WE, Machn D. The seasonal incidence of lichen planus. Trans St Johns Hosp Dermatol Soc. 1968;54(1):69–72.
85. Fessa C, Lim P, Kossard S, Richards S, Peñas PF. Lichen planus-like drug eruptions due to β-blockers: a case report and literature review. Am J Clin Dermatol. 2012;13(6):417–21.
86. Oh SH, Byun HJ, Oh SJ, Jun JY, Park JH, Lee JH, et al. Olmutinib induced lichen planus like eruption. Ann Dermatol. 2018;30(4):451–3.
87. Farah M, Nelson KC, Tetzlaff MT, Nagarajan P, Torres-Cabala CA, Prieto VG, et al. Lichen planus related to transforming growth factor beta inhibitor in a patient with metastatic chondrosarcoma: a case report. J Cutan Pathol. 2020;47(5):490–3.
88. Chan WP, Mackey VT, Sun DK. Telmisartan-induced lichen planus eruption manifested on vitiliginous skin. Cutis. 2017;99(1):E16–9.
89. Ohtani A, Takenaka Y, Ishiguro N, Kawashima M. Case of lichen planus induced by sitagliptin phosphate hydrate. J Dermatol. 2017;44(9):1081–2.
90. Tan SV, Berth-Jones J, Burns DA. Lichen planus and photo-onycholysis induced by quinine. Clin Exp Dermatol. 1989;14(4):335.
91. Dinh H, Seyffert J, Lountzis NI, Altman HB, Oram C, Purcell SM. Lichen planus with predominate plasma cell infiltrate: two case reports. Am J Dermatopathol. 2017;39(2):140–3.
92. Cachia M, Betts A, Aquilina S. Denosumab and lichen planus: two cases. Clin Exp Dermatol. 2021;46(2):356–7.
93. Inoue A, Sawada Y, Yamaguchi T, Ohmori S, Omoto D, Haruyama S, et al. Lichenoid drug eruption caused by adalimumab: a case report and literature review. Eur J Dermatol. 2017;27(1):69–70.
94. Komori T, Honda T, Irie H, Otsuka A, Kabashima K. Lichen planus in irradiated skin during nivolumab treatment. Acta Derm Venereol. 2017;97(3):391–2.
95. Coscarart A, Martel J, Lee MP, Wang AR. Pembrolizumab-induced pseudoepitheliomatous eruption consistent with hypertrophic lichen planus. J Cutan Pathol. 2020;47(3):275–9.
96. Kaunitz G, Kim RH. Lichen planus follicularis tumidus associated with PD-1 blockade. JAMA Dermatol. 2019;155(10):1197–8.
97. de Lorenzi C, André R, Vuilleumier A, Kaya G, Abosaleh M. Bullous lichen planus and anti-programmed cell death-1 therapy: case report and literature review. Ann Dermatol Venereol. 2020;147(3):221–7.
98. Komori T, Honda T, Irie H, Otsuka A, Kabashima K. Multiple erosive lichen planus preceded by solitary lichen planus after combination therapy with nivolumab and radiation. J Eur Acad Dermatol Venereol. 2017;31(8):e382–4.
99. Denny J, Chong H, Akhras V. Lichen planus in a patient treated with pembrolizumab for metastatic malignant melanoma. Clin Exp Dermatol. 2018;43(3):354–6.
100. Liu SS, Howard T, Fattah YH, Adams A, Hanly AJ, Karai LJ. Lichen planopilaris pemphigoides: a novel bullous dermatosis due to programmed cell death protein-1 inhibitor therapy. Am J Dermatopathol. 2023;45(4):246–9.
101. Wang S, Sun J, Deng S, Wu L, Li W, Ye T, et al. Lichen planus pemphigoides induced by anti-PD-1 antibody: A case only involved in oral mucosa with excellent topical treatment efficiency. J Cutan Pathol. 2024;51:114–8.
102. Madan V, Marchitto MC, Sunshine JC. Pembrolizumab-induced lichen planus pemphigoides in a patient with metastatic adrenocortical cancer: a case report and literature review. Dermatopathology (Basel). 2023;10(3):244–58.
103. Boyle MM, Ashi S, Puiu T, Reimer D, Sokumbi O, Soltani K, et al. Lichen planus pemphigoides associated with PD-1 and PD-L1 inhibitors: a case series and review of the literature. Am J Dermatopathol. 2022;44(5):360–7.
104. Shengyuan L, Songpo Y, Wen W, Wenjing T, Haitao Z, Binyou W. Hepatitis C virus and lichen planus: a reciprocal

association determined by a meta-analysis. Arch Dermatol. 2009;145(9):1040–7.
105. Lodi G, Pellicano R, Carrozzo M. Hepatitis C virus infection and lichen planus: a systematic review with meta-analysis. Oral Dis. 2010;16(7):601–12.
106. Ansari U, Henderson LI, Stott G, Parr K. Treatment with ledipasvir-sofosbuvir for hepatitis C resulting in improvement of lichen planus. JAAD Case Rep. 2017;3(1):67–9.
107. Barajas L, Steuer M, Amerson E. Variable clinical course of lichen planus following hepatitis C cure with direct-acting antivirals: a case series and literature review. J Am Acad Dermatol. 2021;84(4):1142–6.
108. Morgado-Carrasco D, Combalia A, Fustà-Novell X, Mascaró JM Jr, Iranzo P. Aggressive erosive lichen planus associated with hepatitis C responding to sofosbuvir/ledipasvir treatment. Indian J Dermatol Venereol Leprol. 2019;85(3):326–9.
109. Wang JH, Hung SJ. Lichen planus associated with hepatitis B, hepatitis C, and liver cirrhosis in a nationwide cohort study. J Am Acad Dermatol. 2021;84(4):1085–6.
110. Dogan B. Dermatological manifestations in hepatitis B surface antigen carriers in east region of Turkey. J Eur Acad Dermatol Venereol. 2005;19(3):323–5.
111. González-Moles M, de Porras-Carrique T, Ramos-García P. Association of oral lichen planus with hepatic disorders and hepatocellular carcinoma: systematic review and meta-analysis. Med Oral Patol Oral Cir Bucal. 2023;28(3):e229–37.
112. Feschuk AM, Green M, Kashetsky N, Maibach HI. Lichen planus following SARS-CoV-2 infection and vaccination? A systematic review Int J Dermatol. 2023;62(2):e54–6.
113. Torre K, Grant-Kels JM, Oliviero M, Rabinovitz H. Lichen planus eruption following annual influenza vaccination. JAAD Case Rep. 2021;10:116–8.
114. Bombeccari GP, Giannì AB, Spadari F. Oral Candida colonization and oral lichen planus. Oral Dis. 2017;23(7):1009–10.
115. Liu J, Geng F, Sun H, Wang X, Zhang H, Yang Q, et al. Candida albicans induces TLR2/MyD88/NF-κB signaling and inflammation in oral lichen planus-derived keratinocytes. J Infect Dev Ctries. 2018;12(9):780–6.
116. Solimani F, Pollmann R, Schmidt T, Schmidt A, Zheng X, Savai R, et al. Therapeutic targeting of Th17/Tc17 cells leads to clinical improvement of lichen planus. Front Immunol. 2019;10:1808.
117. Syrjänen S, Lodi G, von Bültzingslöwen I, Aliko A, Arduino P, Campisi G, et al. Human papillomaviruses in oral carcinoma and oral potentially malignant disorders: a systematic review. Oral Dis. 2011;17(Suppl 1):58–72.
118. de Vries HJ, Teunissen MB, Zorgdrager F, Picavet D, Cornelissen M. Lichen planus remission is associated with a decrease of human herpes virus type 7 protein expression in plasmacytoid dendritic cells. Arch Dermatol Res. 2007;299(4):213–9.
119. De Vries HJ, van Marle J, Teunissen MB, Picavet D, Zorgdrager F, Bos JD, et al. Lichen planus is associated with human herpesvirus type 7 replication and infiltration of plasmacytoid dendritic cells. Br J Dermatol. 2006;154(2):361–4.
120. Karimova M, Moyes D, Ide M, Setterfield JF. The human microbiome in immunobullous disorders and lichen planus. Clin Exp Dermatol. 2022;47(3):522–8.
121. Ertugrul AS, Arslan U, Dursun R, Hakki SS. Periodontopathogen profile of healthy and oral lichen planus patients with gingivitis or periodontitis. Int J Oral Sci. 2013;5(2):92–7.
122. Yiannias JA, El-Azhary RA, Hand JH, Pakzad SY, Rogers RS 3rd. Relevant contact sensitivities in patients with the diagnosis of oral lichen planus. J Am Acad Dermatol. 2000;42(21):177–82.
123. McParland H, Warnakulasuriya S. Oral lichenoid contact lesions to mercury and dental amalgam–a review. J Biomed Biotechnol. 2012;2012: 589569.
124. Agha-Hosseini F, Gholamrezayi E, Moosavi M-S. Patch test of dental materials in Oral Lichen Planus with considering the role of saliva. Sci Rep. 2021;11(1):8264.
125. Quist SR, Papakonstantinou E, Ambach A, Quist J, Göppner D, Reinhold A, et al. Verrucous lichen planus following contact sensitivity to implanted gentamicin-polymethylmethacrylate bead chains. J Eur Acad Dermatol Venereol. 2017;31(1):e35–6.
126. Sasaki G, Yokozeki H, Katayama I, Nishioka K. Three cases of linear lichen planus caused by dental metal compounds. J Dermatol. 1996;23(12):890–2.
127. Sagi L, Trau H. The Koebner phenomenon. Clin Dermatol. 2011;29(2):231–6.
128. Svigos K, Yin L, Fried L, Lo Sicco K, Shapiro J. A practical approach to the diagnosis and management of classic lichen planopilaris. Am J Clin Dermatol. 2021;22(5):681–92.
129. Chiang YZ, Tosti A, Chaudhry IH, Lyne L, Farjo B, Farjo N, et al. Lichen planopilaris following hair transplantation and facelift surgery. Br J Dermatol. 2012;166(3):666–370.
130. Eisen D. The clinical features, malignant potential, and systemic associations of oral lichen planus: a study of 723 patients. J Am Acad Dermatol. 2002;46(2):207–14.
131. Kluger N. Radiation-associated lichen planus: a case report and literature review. Acta Dermatovenerol Alp Pannonica Adriat. 2017;26(4):105–8.
132. Hadian Y, Chen YC, Eastham DV, Schulman JM, Sood A. Radiation induced lichen planus-an uncommon side effect. Dermatol Online J. 2019;25(11):1–6.
133. Cirotteau P, Lucas P, Dousset L, Cales V, Gilliot O, Jouary T. Radiation-induced bullous lichen planus: an isomorphic manifestation of radiotherapy. Acta Derm Venereol. 2019;99(11):1033–4.
134. Melin A, Bouchereau S, Guelimi R, Bergqvist C, Condamina M, Giraud-Kerleroux L, et al. Safety and efficacy of high-dose clobetasol propionate 0.05% in cutaneous lichen planus. J Dermatol. 2023;50(6):810–3.
135. Alchorne MMA, Conceição KDC, Barraza LL, Milanez Morgado de Abreu MA. Dermatology in black skin. An Bras Dermatol. 2024;99(3):327–41.
136. Zaidi Z. Skin of colour: characteristics and disease. J Pak Med Assoc. 2017;67(2):292–9.
137. Munera-Campos M, Castillo G, Ferrándiz C, Carrascosa JM. Actinic lichen planus triggered by drug photosensitivity. Photodermatol Photoimmunol Photomed. 2019;35(2):124–6.
138. Murao K, Tamaki M, Kinoshita H, Kubo Y. Case of actinic lichen planus: the second Japanese case. J Dermatol. 2019;46(8):e304–5.
139. Frioui R, Mokni S, Tabka M, Ounallah A, Ouni NEI, Fetoui N, et al. Inverse lichen planus and palmoplantar keratoderma induced by imatinib mesylate. Int J Dermatol. 2023;62(1):e44–5.
140. Merio L, Tounkara TM, Battesti G, Cordoliani F, Arsouze A, Bagot M, et al. Blaschko-linear lichen planus of the face: a retrospective study of 6 cases and a literature review. Ann Dermatol Venereol. 2022;149(2):112–8.
141. Criscito MC, Brinster NK, Skopicki DL, Seidenberg R, Cohen JM. Blaschkoid lichen planus: Throwing a "curve" in the nomenclature of linear lichen planus. JAAD Case Rep. 2020;6(3):237–9.
142. Zhou Y, Yu ZZ, Peng J, Yang PP, Li SJ, Fan YM. Lichen striatus versus linear lichen planus: a comparison of clinicopathological features, immunoprofile of infiltrated cells, and epidermal proliferation and differentiation. Int J Dermatol. 2016;55(4):e204–10.

143. Manjunath S, Kamat D, Kumaran MS. Multidermatomal zosteriform lichen planus pigmentosus-a unique presentation. Int J Dermatol. 2019;58(3):343–4.
144. Feily A, Yaghoobi R, Nilforoushzadeh MA. Treatment modalities of palmoplantar lichen planus: a brief review. Postepy Dermatol Alergol. 2016;33(6):411–5.
145. Marnach ML, Torgerson RR. Therapeutic interventions for challenging cases of vulvar lichen sclerosus and lichen planus. Obstet Gynecol. 2021;138(3):374–8.
146. Santhosh P, George M. Annular atrophic lichen planus: a review of the literature. Int J Dermatol. 2022;61(10):1213–7.
147. Krispinsky AJ, Ulman CA. Atrophic lichen planus. J Clin Aesthet Dermatol. 2019;12(10):14.
148. Chang HC, Sung CW, Lin MH. Association of circumcision status with genital lichen planus: a systematic review and meta-analysis. Acta Derm Venereol. 2019;99(11):1049–50.
149. Kashima A, Tajiri A, Yamashita A, Asada Y, Setoyama M. Two Japanese cases of lichen planus pigmentosus-inversus. Int J Dermatol. 2007;46(7):740–2.
150. Robles-Méndez JC, Rizo-Frías P, Herz-Ruelas ME, Pandya AG, Ocampo CJ. Lichen planus pigmentosus and its variants: review and update. Int J Dermatol. 2018;57(5):505–14.
151. Dabas G, Vinay K, Parsad D, Chatterjee D, Kumaran MS. A retrospective study of lichen planus pigmentosus with focus on palmoplantar involvement. Clin Exp Dermatol. 2019;44(2):190–3.
152. Kumarasinghe SPW, Pandya A, Chandran V, Rodrigues M, Dlova NC, Kang HY, et al. A global consensus statement on ashy dermatosis, erythema dyschromicum perstans, lichen planus pigmentosus, idiopathic eruptive macular pigmentation, and Riehl's melanosis. Int J Dermatol. 2019;58(3):263–72.
153. Kumaran MS, Dabas G, Parsad D, Vinay K. Lichen planus pigmentosus—an appraisal. Int J Dermatol. 2018;57(6):748–50.
154. Vinay K, Kumar S, Bishnoi A, Aggarwal D, Radotra BD, Parsad D, et al. A clinico-demographic study of 344 patients with lichen planus pigmentosus seen in a tertiary care center in India over an 8-year period. Int J Dermatol. 2020;59(2):245–52.
155. Arnold D, Hoffman MB, Onajin O, Sangüeza OP, McMichael A. Hypopigmented macules as manifestation of lichen planus and lichen planopilaris. Am J Dermatopathol. 2019;41(7):514–7.
156. Reich HL, Nguyen JT, James WD. Annular lichen planus: a case series of 20 patients. J Am Acad Dermatol. 2004;50(4):595–9.
157. Chen X, Wang S. Case of erythrodermic lichen planus successful treated with thalidomide monotherapy. J Dermatol. 2020;47(6):e224–5.
158. Cram DL, Muller SA. Unusual variations of lichen planus. Mayo Clin Proc. 1966;41(10):677–88.
159. Tripathy DM, Vashisht D, Rathore G, Sengupta P. Bullous lichen planus vs lichen planus pemphigoides: a diagnostic dilemma. Indian Dermatol Online J. 2022;13(2):282–4.
160. Scotti B, Veronesi G, Misciali C, Venturi F, Dika E. Unusual presentation of porokeratotic lichen planus: Histology, dermoscopy and confocal microscopy imaging of a rare condition. Skin Res Technol. 2023;29(11): e13521.
161. Dhanta A, Kansal NK, Durgapal P, Divyalakshmi C. Porokeratotic lichen planus. J Dtsch Dermatol Ges. 2019;17(10):1063–5.
162. Tekin B, Johnson EF, Wieland CN, Gibson LE, Camilleri MJ, Kalaaji AN, et al. Histopathology of autoimmune bullous dermatoses: what's new? Hum Pathol. 2022;128:69–89.
163. Matos-Pires E, Campos S, Lencastre A, João A, Mendes-Bastos P. Lichen planus pemphigoides. J Dtsch Dermatol Ges. 2018;16(3):335–7.
164. Hübner F, Langan EA, Recke A. Lichen planus pemphigoides: from lichenoid inflammation to autoantibody-mediated blistering. Front Immunol. 2019;10:1389.
165. Ondhia C, Kaur C, Mee J, Natkunarajah J, Singh M. Lichen planus pemphigoides mimicking toxic epidermal necrolysis. Am J Dermatopathol. 2019;41(11):e144–7.
166. Loyal J, Rashtak S. Vulvar lichen planus pemphigoides. Int J Womens Dermatol. 2017;3(4):225–7.
167. Lamberts A, Diercks GFH, Pas HH, Horváth B. Non-bullous lichen planus pemphigoides: a case report. Acta Derm Venereol. 2020;100(10):adv00156.
168. Knisley RR, Petropolis AA, Mackey VT. Lichen planus pemphigoides treated with ustekinumab. Cutis. 2017;100(6):415–8.
169. Lospinoso D, Fernelius C, Edhegard K, Finger D, Arora N. Lupus erythematosus/lichen planus overlap syndrome: successful treatment with acitretin. Lupus. 2013;22(8):851–4.
170. Komori T, Otsuka A, Honda T, Kaku Y, Kabashima K. A case of chilblain lupus erythematosus with lupus erythematosus/lichen planus overlap syndrome. J Eur Acad Dermatol Venereol. 2017;31(9):e424–5.
171. Nagao K, Chen KR. A case of lupus erythematosus/lichen planus overlap syndrome. J Dermatol. 2006;33(3):187–90.
172. Schmitz S, Vatanchi M, Alapati U. Seven-year itch: a perplexing case of lichen planus-lupus erythematosus overlap syndrome. Dermatol Online J. 2018;24(9).
173. Dunaway S, Tyler K, Kaffenberger J. Update on treatments for erosive vulvovaginal lichen planus. Int J Dermatol. 2020;59(3):297–302.
174. Chao TJ. Adalimumab in the management of cutaneous and oral lichen planus. Cutis. 2009;84(6):325–8.
175. Didona D, Caposiena Caro RD, Sequeira Santos AM, Solimani F, Hertl M. Therapeutic strategies for oral lichen planus: State of the art and new insights. Front Med (Lausanne). 2022;9: 997190.
176. Moussa A, Colla T, Morrison B, Sinclair R. Effective treatment of oral lichen planus with the JAK inhibitor baricitinib. Australas J Dermatol. 2022;63(2):276–7.
177. AbuHilal M, Walsh S, Shear N. Treatment of recalcitrant erosive oral lichen planus and desquamative gingivitis with oral apremilast. J Dermatol Case Rep. 2016;10(3):56–7.
178. van Serooskerken AM, van Marion AM, de Zwart-Storm E, Frank J, Poblete-Gutiérrez P. Lichen planus with bullous manifestation on the lip. Int J Dermatol. 2007;46(Suppl 3):25–6.
179. Cooper SM, Wojnarowska F. Influence of treatment of erosive lichen planus of the vulva on its prognosis. Arch Dermatol. 2006;142(3):289–94.
180. Byrd JA, Davis MD, Rogers RS 3rd. Recalcitrant symptomatic vulvar lichen planus: response to topical tacrolimus. Arch Dermatol. 2004;140(6):715–20.
181. Dubey R, Fischer G. Vulvo-vaginal lichen planus: a focussed review for the clinician. Australas J Dermatol. 2019;60(1):7–11.
182. Xie F, Dasari S, Deschaine M, Gleue CA, Sartori-Valinotti JC, Torgerson RR, et al. Differential proteomic expression profiles in vulvar lichen planus as compared to normal vulvar tissue, vulvar lichen sclerosus, or oral lichen planus: an exploratory study. Exp Dermatol. 2023;32(9):1498–508.
183. Day T, Weigner J, Scurry J. Classic and hypertrophic vulvar lichen planus. J Low Genit Tract Dis. 2018;22(4):387–95.
184. Boch K, Langan EA, Zillikens D, Ludwig RJ, Kridin K. Retrospective analysis of the clinical characteristics and patient-reported outcomes in vulval lichen planus: results from a single-center study. J Dermatol. 2021;48(12):1913–7.
185. Kherlopian A, Fischer G. Successful treatment of vulvovaginal lichen planus with tildrakizumab: a case series of 24 patients. Australas J Dermatol. 2022;63(2):251–5.
186. Żychowska M, Woźniak Z, Maj J. Vulvovaginal-gingival-pilar lichen planus in a patient with autoimmune hepatitis. Int J Dermatol. 2018;57(12):1500–1.

187. Rezzag-Mahcene C, Cardot-Leccia N, Lacour JP, Montaudié H, Passeron T. Successful treatment of recalcitrant genital lichen planus with secukinumab. J Eur Acad Dermatol Venereol. 2021;35(5):e321–3.
188. Kraus CN, Smith J. The many faces of penile lichen planus. Int J Dermatol. 2019;58(12):1460–1.
189. Amsellem J, Skayem C, Duong TA, Bagot M, Fouéré S, Dauendorffer JN. Male genital lichen planus: a retrospective study of 89 cases. Ann Dermatol Venereol. 2022;149(1):28–31.
190. Decker A, Schauer F, Lazaro A, Monasterio C, Schmidt AR, Schmitt-Graeff A, et al. Esophageal lichen planus: Current knowledge, challenges and future perspectives. World J Gastroenterol. 2022;28(41):5893–909.
191. Kooybaran NR, Petzold G, Schön MP, Mössner R. Esophageal lichen planus successfully treated with the JAK1/3 inhibitor tofacitinib. J Dtsch Dermatol Ges. 2022;20(6):858–60.
192. Schauer F, Monasterio C, Technau-Hafsi K, Kern JS, Lazaro A, Deibert P, et al. Esophageal lichen planus: towards diagnosis of an underdiagnosed disease. Scand J Gastroenterol. 2019;54(10):1189–98.
193. Ozlu E, Teberik K. Evaluation of ocular findings in patients with lichen planus. Postepy Dermatol Alergol. 2019;36(3):267–71.
194. Altunisik N, Koc A, Toplu Y, Sener S, Turkmen D, Sarac G, et al. Evaluation of hearing function in patients with lichen planus. Med Hypotheses. 2020;143: 109867.
195. Sartori-Valinotti JC, Bruce AJ, Krotova Khan Y, Beatty CW. A 10-year review of otic lichen planus: the Mayo Clinic experience. JAMA Dermatol. 2013;149(9):1082–6.
196. Sharma N, Malhotra SK, Kuthial M, Chahal KS. Vulvo-vaginal ano-gingival syndrome: Another variant of mucosal lichen planus. Indian J Sex Transm Dis AIDS. 2017;38(1):86–8.
197. Lajevardi V, Ghodsi SZ, Goodarzi A, Hejazi P, Azizpour A, Beygi S. Comparison of systemic mycophenolate mofetil with topical clobetasol in lichen planopilaris: a parallel-group, assessor- and analyst-blinded, randomized controlled trial. Am J Clin Dermatol. 2015;16(4):303–11.
198. Manatis-Lornell A, Okhovat JP, Marks DH, Hagigeorges D, Senna MM. Comorbidities in patients with lichen planopilaris: a retrospective case-control study. J Am Acad Dermatol. 2020;83(1):205–8.
199. Cantwell HM, Wieland CN, Proffer SL, Imhof RL, Torgerson RR, Tolkachjov SN. Lichen planopilaris in men: a retrospective clinicopathologic study of 19 patients. Int J Dermatol. 2021;60(4):482–8.
200. Xie F, Lehman JS. Lichen Planopilaris. Mayo Clin Proc. 2022;97(2):208–9.
201. Cevasco NC, Bergfeld WF, Remzi BK, de Knott HR. A case-series of 29 patients with lichen planopilaris: the Cleveland Clinic Foundation experience on evaluation, diagnosis, and treatment. J Am Acad Dermatol. 2007;57(1):47–53.
202. Jordan CS, Chapman C, Kolivras A, Roberts JL, Thompson NB, Thompson CT. Clinicopathologic and immunophenotypic characterization of lichen planopilaris and central centrifugal cicatricial alopecia: a comparative study of 51 cases. J Cutan Pathol. 2020;47(2):128–34.
203. Griffiths AAC, Sancho MI, Plaza AI. Piccardi-Lassueur-Graham-Little syndrome associated with frontal fibrosing alopecia. An Bras Dermatol. 2017;92(6):867–9.
204. Gálvez-Canseco A, Sperling L. Lichen planopilaris and frontal fibrosing alopecia cannot be differentiated by histopathology. J Cutan Pathol. 2018;45(5):313–7.
205. Munck A, Valente N, Romiti R. Axillary involvement in frontal fibrosing alopecia: clinical, dermoscopy and histological features. J Eur Acad Dermatol Venereol. 2019;33(4):e160–71.
206. Pedrosa AF, Duarte AF, Haneke E, Correia O. Yellow facial papules associated with frontal fibrosing alopecia: a distinct histologic pattern and response to isotretinoin. J Am Acad Dermatol. 2017;77(4):764–6.
207. Vañó-Galván S, Saceda-Corralo D, Moreno-Arrones ÓM, Camacho-Martinez FM. Updated diagnostic criteria for frontal fibrosing alopecia. J Am Acad Dermatol. 2018;78(1):e21–2.
208. Imhof RL, Chaudhry HM, Larkin SC, Torgerson RR, Tolkachjov SN. Frontal fibrosing alopecia in women: the mayo clinic experience with 148 patients, 1992–2016. Mayo Clin Proc. 2018;93(11):1581–8.
209. Westphal DC, Caballero-Uribe N, Regnier A, Taguti P, Dutra Rezende H, Trüeb RM. Male frontal fibrosing alopecia: study of 35 cases and association with sunscreens, facial skin and hair care products. J Eur Acad Dermatol Venereol. 2021;35(9):e587–9.
210. Maghfour J, Ceresnie M, Olson J, Lim HW. The association between frontal fibrosing alopecia, sunscreen, and moisturizers: a systematic review and meta-analysis. J Am Acad Dermatol. 2022;87(2):395–6.
211. Aldoori N, Dobson K, Holden CR, McDonagh AJ, Harries M, Messenger AG. Frontal fibrosing alopecia: possible association with leave-on facial skin care products and sunscreens; a questionnaire study. Br J Dermatol. 2016;175(4):762–7.
212. Felmingham C, Yip L, Tam M, Nixon RL. Allergy to sunscreen and leave-on facial products is not a likely causative mechanism in frontal fibrosing alopecia: perspective from contact allergy experts. Br J Dermatol. 2020;182(2):481–2.
213. Żychowska M, Żychowska M. Nail changes in lichen planus: a single-center study. J Cutan Med Surg. 2021;25(3):281–5.
214. Kharghoria G, Grover C, Bhattacharya SN, Sharma S. Histopathological evaluation of nail lichen planus: a cross-sectional study. J Cutan Pathol. 2021;48(1):11–7.
215. Iorizzo M, Tosti A, Starace M, Baran R, Daniel CR 3rd, Di Chiacchio N, et al. Isolated nail lichen planus: An expert consensus on treatment of the classical form. J Am Acad Dermatol. 2020;83(6):1717–23.
216. Pünchera J, Laffitte E. Treatment of severe nail lichen planus with baricitinib. JAMA Dermatol. 2022;158(1):107–8.
217. McClanahan DR, English JC 3rd. Therapeutics for adult nail psoriasis and nail lichen planus: a guide for clinicians. Am J Clin Dermatol. 2018;19(4):559–84.
218. Iorizzo M. Nail lichen planus - a possible new indication for oral alitretinoin. J Eur Acad Dermatol Venereol. 2016;30(3):509–10.
219. Chessa MA, Alessandrini A, Starace M, Baraldi C, Dahdah M, Andre J, et al. Erosive lichen planus: beyond the nails. J Eur Acad Dermatol Venereol. 2019;33(3):e97–9.
220. Tosti A, Piraccini BM, Cameli N. Nail changes in lichen planus may resemble those of yellow nail syndrome. Br J Dermatol. 2000;142(4):848–9.
221. Kassels A, Elsensohn A, Kraus CN. Lichen planus is associated with other autoimmune conditions: a retrospective population-level study. J Am Acad Dermatol. 2024;90(3):650–2.
222. Fromme M, Schneider CV, Schlapbach C, Cazzaniga S, Trautwein C, Rader DJ, et al. Comorbidities in lichen planus by phenome-wide association study in two biobank population cohorts. Br J Dermatol. 2022;187(5):722–9.
223. Fan R, Leasure AC, Cohen JM. Association of autoimmune comorbidities with lichen planus: a United States-based case-control study in the All of Us research program. J Am Acad Dermatol. 2022;87(6):1451–3.
224. De Porras-Carrique T, Ramos-García P, Aguilar-Diosdado M, Warnakulasuriya S, González-Moles M. Autoimmune disorders in oral lichen planus: a systematic review and meta-analysis. Oral Dis. 2023;29(4):1382–94.

225. Gibson LE, Muller SA. Dermatologic disorders in patients with thymoma. Acta Derm Venereol. 1987;67(4):351–6.
226. Hayashi A, Shiono H, Okumura M. Thymoma accompanied by lichen planus. Interact Cardiovasc Thorac Surg. 2008;7(2):347–8.
227. Teramura K, Kato T, Nishikawa J, Nakanishi T, Tanaka T, Fujimoto N. Oral lichen planus and lichen planopilaris complicated with thymoma. J Dermatol. 2019;46(7):e237–9.
228. Khan S, Campbell A, Hunt C, Sewell WA. Lichen planus in a case of Good's syndrome (thymoma and immunodeficiency). Interact Cardiovasc Thorac Surg. 2009;9(2):345–6.
229. Lolis M, Levitt J. A case of oral lichen planus with Good syndrome. Dermatol Online J. 2006;12(6):28.
230. Nasiri S, Sadeghzadeh-Bazargan A, Robati RM, Haghighatkhah HR, Younespour S. Subclinical atherosclerosis and cardiovascular markers in patients with lichen planus: a case-control study. Indian J Dermatol Venereol Leprol. 2019;85(2):138–44.
231. Aksu F, Karadag AS, Caliskan M, Uzuncakmak TK, Keles N, Ozlu E, et al. Does lichen planus cause increased carotid intima-media thickness and impaired endothelial function? Can J Cardiol. 2016;32(10):12461.e1-e6.
232. Leasure AC, Acosta JN, Sansing LH, Sheth KN, Cohen JM, Falcone GJ. Association of lichen planus with cardiovascular disease: A combined analysis of the UK Biobank and All of Us Study. J Am Acad Dermatol. 2022;87(2):454–6.
233. Özkur E, Uğurer E, Altunay İK. Dyslipidemia in lichen planus: a case-control study. Sisli Etfal Hastan Tip Bul. 2020;54(1):62–6.
234. Ertem AG, Erdogan M, Koseoglu C, Akoglu G, Ozdemir E, Koseoglu G, et al. Epicardial fat tissue thickness is increased in patients with lichen planus and is linked to inflammation and dyslipidemia. Rev Port Cardiol. 2016;35(10):525–30.
235. Baykal L, Arıca DA, Yaylı S, Örem A, Bahadır S, Altun E, et al. Prevalence of metabolic syndrome in patients with mucosal lichen planus: a case-control study. Am J Clin Dermatol. 2015;16(5):439–45.
236. Arias-Santiago S, Buendía-Eisman A, Aneiros-Fernández J, Girón-Prieto MS, Gutiérrez-Salmerón MT, Mellado VG, et al. Cardiovascular risk factors in patients with lichen planus. Am J Med. 2011;124(6):543–8.
237. Otero Rey EM, Yáñez-Busto A, Rosa Henriques IF, López-López J, Blanco-Carrión A. Lichen planus and diabetes mellitus: Systematic review and meta-analysis. Oral Dis. 2019;25(5):1253–64.
238. López-Jornet P, Camacho-Alonso F, Rodríguez-Martínes MA. Alterations in serum lipid profile patterns in oral lichen planus: a cross-sectional study. Am J Clin Dermatol. 2012;13(6):399–404.
239. Arias-Santiago S, Buendía-Eisman A, Aneiros-Fernández J, Girón-Prieto MS, Gutiérrez-Salmerón MT, García-Mellado V, et al. Lipid levels in patients with lichen planus: a case-control study. J Eur Acad Dermatol Venereol. 2011;25(12):1398–401.
240. Ying J, Xiang W, Qiu Y, Zeng X. Risk of metabolic syndrome in patients with lichen planus: a systematic review and meta-analysis. PLoS ONE. 2020;15(8): e0238005.
241. Lai YC, Yew YW, Schwartz RA. Lichen planus and dyslipidemia: a systematic review and meta-analysis of observational studies. Int J Dermatol. 2016;55(5):e295-304.
242. Fertig RM, Hu S, Maddy AJ, Balaban A, Aleid N, Aldahan A, et al. Medical comorbidities in patients with lichen planopilaris, a retrospective case-control study. Int J Dermatol. 2018;57(7):804–9.
243. Conic RRZ, Piliang M, Bergfeld W, Atanaskova-Mesinkovska N. Association of lichen planopilaris with dyslipidemia. JAMA Dermatol. 2018;154(9):1088–9.
244. Phan K, Smith SD. Lichen planopilaris and dyslipidaemia: systematic review and meta-analysis. Clin Exp Dermatol. 2020;45(5):611–3.
245. Sigurgeirsson B, Lindelöf B. Lichen planus and malignancy. An epidemiologic study of 2071 patients and a review of the literature. Arch Dermatol. 1991;127(11):1684–8.
246. Gonzalez-Moles MA, Scully C, Gil-Montoya JA. Oral lichen planus: controversies surrounding malignant transformation. Oral Dis. 2008;14(3):229–43.
247. González-Moles M, Ruiz-Ávila I, González-Ruiz L, Ayén Á, Gil-Montoya JA, Ramos-García P. Malignant transformation risk of oral lichen planus: a systematic review and comprehensive meta-analysis. Oral Oncol. 2019;96:121–30.
248. Ghandi N, Sadeghi M, Kalantari Y, Ghodsi SZ, Mahmoudi H, Etesami I. Squamous cell carcinoma (SCC) and intraepithelial neoplasia among patients diagnosed with erosive lichen planus (LP): A study of 130 patients. Australas J Dermatol. 2023:64(4):546–8.
249. Laniosz V, Torgerson RR, Ramos-Rodriguez AJ, Ma JE, Mara KC, Weaver AL, et al. Incidence of squamous cell carcinoma in oral lichen planus: a 25-year population-based study. Int J Dermatol. 2019;58(3):296–301.
250. González-Moles M, Warnakulasuriya S, González-Ruiz I, González-Ruiz L, Ayén Á, Lenouvel D, et al. Clinicopathological and prognostic characteristics of oral squamous cell carcinomas arising in patients with oral lichen planus: a systematic review and a comprehensive meta-analysis. Oral Oncol. 2020;106: 104688.
251. Warnakulasuriya S, Kujan O, Aguirre-Urizar JM, Bagan JV, González-Moles M, Kerr AR, et al. Oral potentially malignant disorders: a consensus report from an international seminar on nomenclature and classification, convened by the WHO Collaborating Centre for Oral Cancer. Oral Dis. 2021;27(8):1862–80.
252. !!! INVALID CITATION !!! [247].
253. Idrees M, Kujan O, Shearston K, Farah CS. Oral lichen planus has a very low malignant transformation rate: A systematic review and meta-analysis using strict diagnostic and inclusion criteria. J Oral Pathol Med. 2021;50(3):287–98.
254. Xie F, Meves A, Lehman JS. The genomic and proteomic landscape in oral lichen planus versus oral squamous cell carcinoma: a scoping review. Int J Dermatol. 2022;61(10):1227–36.
255. Xie F, Gleue CA, Deschaine M, Dasari S, Lau JS, Sartori-Valinotti JC, et al. Whole-exome sequencing of transforming oral lichen planus reveals mutations in DNA damage repair and apoptosis pathway genes. J Oral Pathol Med. 2022;51(4):395–404.
256. Kherlopian A, Fischer G. Vulvar malignancy in biopsy-proven vulval lichen planus: a retrospective review of 105 cases. Australas J Dermatol. 2020;61(4):386–8.
257. Kristiansen S, Svensson Å, Drevin L, Forslund O, Torbrand C, Bjartling C. Risk factors for penile intraepithelial neoplasia: a population-based register study in Sweden, 2000–2012. Acta Derm Venereol. 2019;99(3):315–20.
258. de Oliveira Leal ML, Alencar LRPJ, Santana SC, de Souza BCA, Athanazio DA. Penile squamous cell carcinoma and lichen planus. Surg Exp Pathol. 2020;3(1):1.
259. Hoshi A, Usui Y, Terachi T. Penile carcinoma originating from lichen planus on glans penis. Urology. 2008;71(5):816–7.
260. Ravi K, Codipilly DC, Sunjaya D, Fang H, Arora AS, Katzka DA. Esophageal lichen planus is associated with a significant increase in risk of squamous cell carcinoma. Clin Gastroenterol Hepatol. 2019;17(9):1902-3.e1.
261. Haenen CCP, Buurma AAJ, Genders RE, Quint KD. Squamous cell carcinoma arising in hypertrophic lichen planus. BMJ Case Rep. 2018;11:2018.
262. Knackstedt TJ, Collins LK, Li Z, Yan S, Samie FH. Squamous cell carcinoma arising in hypertrophic lichen planus: a review and analysis of 38 cases. Dermatol Surg. 2015;41(12):1411–8.

263. Totonchy MB, Leventhal JS, Ko CJ, Leffell DJ. Hypertrophic lichen planus and well-differentiated squamous cell carcinoma: a diagnostic conundrum. Dermatol Surg. 2018;44(11):1466–70.
264. Brauns B, Stahl M, Schön MP, Zutt M. Intralesional steroid injection alleviates nail lichen planus. Int J Dermatol. 2011;50(5):626–7.
265. Mazzeo M, Saggini A, Rocco T, Fargnoli MC, Cerroni L. Syringotropic lichen planus: a potential histopathologic mimicker of syringotropic mycosis fungoides. Am J Dermatopathol. 2019;41(5):e50–3.
266. Sanchez I, Ibraheim MK, Lee BA, Kraus CN, Elsensohn A. Eosinophils in traditionally noneosinophil-rich dermatoses. Am J Dermatopathol. 2023:45(12):820–1.
267. Tang MB, Yosipovitch G, Tan SH. Secondary syphilis presenting as a lichen planus-like rash. J Eur Acad Dermatol Venereol. 2004;18(2):185–7.
268. Trinh NB, Wu YH, Hieu HT. Penile annular secondary syphilis mimicking annular lichen planus. Int J Dermatol. 2022;61(3):e115–6.
269. Saleh J, Vasicek B, Swan J, Speiser J, Mudaliar K. A rare case of lichen planus-like atopic dermatitis involving the hands. Am J Dermatopathol. 2020;42(3):215–7.
270. Allen HB, Jones NP, Bowen SE. Lichenoid and other clinical presentations of atopic dermatitis in an inner city practice. J Am Acad Dermatol. 2008;58(3):503–4.
271. Kulthanan K, Jiamton S, Varothai S, Pinkaew S, Sutthipinittharm P. Direct immunofluorescence study in patients with lichen planus. Int J Dermatol. 2007;46(12):1237–41.
272. Helander SD, Rogers RS 3rd. The sensitivity and specificity of direct immunofluorescence testing in disorders of mucous membranes. J Am Acad Dermatol. 1994;30(1):65–75.
273. Rujitharanawong C, Tuchinda P, Chularojanamontri L, Pochanapan O, Nuttawong S, Thongprasom K, et al. Direct immunofluorescence staining patterns compared between oral and cutaneous lichen planus. Clin Exp Dermatol. 2022;47(2):389–93.
274. Yamanaka Y, Yamashita M, Innocentini LMA, Macedo LD, Chahud F, Ribeiro-Silva A, et al. Direct immunofluorescence as a helpful tool for the differential diagnosis of oral lichen planus and oral lichenoid lesions. Am J Dermatopathol. 2018;40(7):491–7.
275. Kerkemeyer KL, Pinczewski J, Sinclair R. Successful treatment of recalcitrant lichen planus pemphigoides with tildrakizumab. Australas J Dermatol. 2020;61(3):e366–8.
276. Litaiem N, Mansour Y, Jones M, Zeglaoui F. Dermoscopic signs of lichen planus. BMJ Case Rep. 2016;11:2016.
277. García-García B, Munguía-Calzada P, Aubán-Pariente J, Argenziano G, Vázquez-López F. Dermoscopy of lichen planus: Vascular and Wickham striae variations in the skin of colour. Australas J Dermatol. 2019;60(4):301–4.
278. Ankad BS, Beergouder SL. Hypertrophic lichen planus versus prurigo nodularis: a dermoscopic perspective. Dermatol Pract Concept. 2016;6(2):9–15.
279. Hanumaiah B, Joseph JM. Role of dermoscopy in the diagnosis of hypertrophic lichen planus and prurigo nodularis. Indian J Dermatol. 2019;64(5):341–5.
280. van der Meij EH, van der Waal I. Lack of clinicopathologic correlation in the diagnosis of oral lichen planus based on the presently available diagnostic criteria and suggestions for modifications. J Oral Pathol Med. 2003;32(9):507–12.
281. Cheng YS, Gould A, Kurago Z, Fantasia J, Muller S. Diagnosis of oral lichen planus: a position paper of the American Academy of Oral and Maxillofacial Pathology. Oral Surg Oral Med Oral Pathol Oral Radiol. 2016;122(3):332–54.
282. Feldmeyer L, Suter VG, Oeschger C, Cazzaniga S, Bornstein MM, Simon D, et al. Oral lichen planus and oral lichenoid lesions - an analysis of clinical and histopathological features. J Eur Acad Dermatol Venereol. 2020;34(2):e104–7.
283. Wu M, Lee G, Fischer G. Forming diagnostic criteria for vulvar lichen planus. Australas J Dermatol. 2020;61(4):324–9.
284. Fruchter R, Melnick L, Pomeranz MK. Lichenoid vulvar disease: a review. Int J Womens Dermatol. 2017;3(1):58–64.
285. Spencer LA, Hawryluk EB, English JC 3rd. Lichen planopilaris: retrospective study and stepwise therapeutic approach. Arch Dermatol. 2009;145(3):333–4.
286. Uchiyama M, Harada K, Tobita R, Irisawa R, Tsuboi R. Histopathologic and dermoscopic features of 42 cases of folliculitis decalvans: a case series. J Am Acad Dermatol. 2021;85(5):1185–93.
287. Egger A, Stojadinovic O, Miteva M. Folliculitis decalvans and lichen planopilaris phenotypic spectrum-a series of 7 new cases with focus on histopathology. Am J Dermatopathol. 2020;42(3):173–7.
288. Melián-Olivera A, Moreno-Arrones Ó, Burgos-Blasco P, Hermosa-Gelbard Á, Jaén-Olasolo P, Vañó-Galván S, et al. Clinical characterization and treatment response of folliculitis decalvans lichen planopilaris phenotypic spectrum: a unicentre retrospective series of 31 patients. Acta Derm Venereol. 2024;104:adv12373.
289. Sharma S, Kaur J, Bassi R, Sekhon HK. Dermatoscopy of nail lichen planus: a boon for early diagnosis. J Clin Aesthet Dermatol. 2021;14(7):34–7.
290. Xie F, Lehman JS. Nail lichen planus. Mayo Clin Proc. 2021;96(5):1363–4.
291. Pousti BT, Jin A, Sklovar L, Savage KT, Zhai LL, Mollanazar NK, et al. Dupilumab for the treatment of lichen planus. Cutis. 2021;107(4):E8-e10.
292. Iorizzo M, Haneke E. Tofacitinib as treatment for nail lichen planus associated with alopecia universalis. JAMA Dermatol. 2021;157(3):352–3.
293. Thandar Y, Maharajh R, Haffejee F, Mosam A. Treatment of cutaneous lichen planus (part 2): a review of systemic therapies. J Dermatolog Treat. 2019;30(7):633–47.
294. Pavlotsky F, Nathansohn N, Kriger G, Shpiro D, Trau H. Ultraviolet-B treatment for cutaneous lichen planus: our experience with 50 patients. Photodermatol Photoimmunol Photomed. 2008;24(2):83–6.
295. Atzmony L, Reiter O, Hodak E, Gdalevich M, Mimouni D. Treatments for cutaneous lichen planus: a systematic review and meta-analysis. Am J Clin Dermatol. 2016;17(1):11–22.
296. Husein-ElAhmed H, Gieler U, Steinhoff M. Lichen planus: a comprehensive evidence-based analysis of medical treatment. J Eur Acad Dermatol Venereol. 2019;33(10):1847–62.
297. Yoo SA, Park HE, Kim M. Alitretinoin for Hypertrophic Lichen Planus. Ann Dermatol. 2022;34(3):235–6.
298. Muthu SK, Narang T, Saikia UN, Kanwar AJ, Parsad D, Dogra S. Low-dose oral isotretinoin therapy in lichen planus pigmentosus: an open-label non-randomized prospective pilot study. Int J Dermatol. 2016;55(9):1048–54.
299. Woo TY. Systemic isotretinoin treatment of oral and cutaneous lichen planus. Cutis. 1985;35(4):385–6 (**90-1, 93**).
300. Rallis E, Liakopoulou A, Christodoulopoulos C, Katoulis A. Successful treatment of bullous lichen planus with acitretin monotherapy. Review of treatment options for bullous lichen planus and case report. J Dermatol Case Rep. 2016;10(4):62–4.
301. Laurberg G, Geiger JM, Hjorth N, Holm P, Hou-Jensen K, Jacobsen KU, et al. Treatment of lichen planus with acitretin. A double-blind, placebo-controlled study in 65 patients. J Am Acad Dermatol. 1991;24(3):434–7.
302. Omidian M, Ayoobi A, Mapar MA, Feily A, Cheraghian B. Efficacy of sulfasalazine in the treatment of generalized lichen planus: randomized double-blinded clinical trial on 52 patients. J Eur Acad Dermatol Venereol. 2010;24(9):1051–4.

303. Sehgal VN, Abraham GJ, Malik GB. Griseofulvin therapy in lichen planus A double-blind controlled trial. Br J Dermatol. 1972;87(4):383–5.
304. Verma KK, Mittal R, Manchanda Y. Azathioprine for the treatment of severe erosive oral and generalized lichen planus. Acta Derm Venereol. 2001;81(5):378–9.
305. Büyük AY, Kavala M. Oral metronidazole treatment of lichen planus. J Am Acad Dermatol. 2000;43(2 Pt 1):260–2.
306. Khandpur S, Sugandhan S, Sharma VK. Pulsed itraconazole therapy in eruptive lichen planus. J Eur Acad Dermatol Venereol. 2009;23(1):98–101.
307. Turan H, Baskan EB, Tunali S, Yazici S, Saricaoglu H. Methotrexate for the treatment of generalized lichen planus. J Am Acad Dermatol. 2009;60(1):164–6.
308. Gamal Alsakaan NA, Abd-Elsalam S, Fawzy MM, Elwan NM. Efficacy and safety of oral methotrexate versus oral mini pulse betamethasone therapy in the treatment of lichen planus: a comparative study. J Dermatolog Treat. 2022;33(7):3039–46.
309. Covington M, Roman C, Abdulla FR. Methotrexate as a treatment of palmoplantar lichen planus. Cutis. 2020;106(2):E5-e7.
310. Hrin ML, Bashyam AM, Huang WW, Feldman SR. Mycophenolate mofetil for the treatment of cutaneous lichen planus: a retrospective case series. J Am Acad Dermatol. 2021;84(4):1091–4.
311. Orozco-Anahuati AP, Morgado-Carrasco D, Fustà-Novell X, Mascaró JM Jr. Lichen planus pemphigoides with complete clinical response after ciclosporin therapy. Australas J Dermatol. 2019;60(3):e254–6.
312. Higgins EM, Munro CS, Friedmann PS, Marks JM. Cyclosporin A in the treatment of lichen planus. Arch Dermatol. 1989;125(10):1436.
313. Cozzani E, Gariazzo L, Cioni M, Parodi A. Could colchicine represent a new therapeutic approach for lichen planus pigmentosus? Dermatol Ther. 2019;32(2): e12809.
314. Paslin DA. Sustained remission of generalized lichen planus induced by cyclophosphamide. Arch Dermatol. 1985;121(2):236–9.
315. Rai R, Kaur I, Kumar B. Low-dose low-molecular-weight heparin in lichen planus. J Am Acad Dermatol. 2002;46(1):141–3.
316. Gharib K, Kandil A, Marie A, Mounir H. Efficacy of intralesional bacillus calmette-guérin polysaccharide nucleic acid and vitamin D injections in the treatment of lichen planus. J Clin Aesthet Dermatol. 2021;14(5):40–5.
317. Cline A, Cuellar-Barboza A, Jorizzo JL, Pichardo RO. Methotrexate for the treatment of recalcitrant erosive lichen planus of the vulva. JAMA Dermatol. 2020;156(2):215–7.
318. Jang N, Fischer G. Treatment of erosive vulvovaginal lichen planus with methotrexate. Australas J Dermatol. 2008;49(4):216–9.
319. Kanwar AJ, De D. Methotrexate for treatment of lichen planus: old drug, new indication. J Eur Acad Dermatol Venereol. 2013;27(3):e410–3.
320. Lajevardi V, Ghodsi SZ, Hallaji Z, Shafiei Z, Aghazadeh N, Akbari Z. Treatment of erosive oral lichen planus with methotrexate. J Dtsch Dermatol Ges. 2016;14(3):286–93.
321. Hrin ML, Bowers NL, Feldman SR, Huang WW, Pichardo RO. Mycophenolate mofetil versus methotrexate for vulvar lichen planus: a 10-year retrospective cohort study demonstrates comparable efficacy and tolerability. J Am Acad Dermatol. 2022;87(2):436–8.
322. Chauhan P, De D, Handa S, Narang T, Saikia UN. A prospective observational study to compare efficacy of topical triamcinolone acetonide 0.1% oral paste, oral methotrexate, and a combination of topical triamcinolone acetonide 0.1% and oral methotrexate in moderate to severe oral lichen planus. Dermatol Ther. 2018;31(1):1–6.
323. Ch'en PY, Song EJ. Lichen planus pemphigoides successfully treated with dupilumab. JAAD Case Rep. 2023;31:56–8.
324. Holló P, Szakonyi J, Kiss D, Jokai H, Horváth A, Kárpáti S. Successful treatment of lichen planus with adalimumab. Acta Derm Venereol. 2012;92(4):385–6.
325. Khattab FM, Samir MA. Assessment of neopterin level and severity in lichen planus patients treated with narrow-band ultraviolet B. J Cosmet Dermatol. 2020;19(12):3389–92.
326. Weber B, Marquart E, Radakovic S, Tanew A. Effectiveness of narrowband UVB phototherapy and psoralen plus UVA photochemotherapy in the treatment of generalized lichen planus: Results from a large retrospective analysis and an update of the literature. Photodermatol Photoimmunol Photomed. 2022;38(2):104–11.
327. Iraji F, Faghihi G, Asilian A, Siadat AH, Larijani FT, Akbari M. Comparison of the narrow band UVB versus systemic corticosteroids in the treatment of lichen planus: a randomized clinical trial. J Res Med Sci. 2011;16(12):1578–82.
328. Dawood M, Sizopoulou C, Greenberger S, Barzilai A, Pavlotsky F. Narrowband ultraviolet B radiation for lichen planus: long-term follow-up of 192 patients. J Clin Aesthet Dermatol. 2022;15(4):31–5.
329. Visconti MJ, Haidari W, Feldman SR. Transcutaneous electrical nerve stimulation (TENS): a review of applications in dermatology. J Dermatolog Treat. 2020;31(8):846–9.
330. Majid I. Fractional carbon dioxide laser in combination with topical corticosteroid: an innovative treatment for hypertrophic lichen planus. J Am Acad Dermatol. 2017;77(3):e67–8.
331. Geisler AN, Phillips GS, Barrios DM, Wu J, Leung DYM, Moy AP, et al. Immune checkpoint inhibitor-related dermatologic adverse events. J Am Acad Dermatol. 2020;83(5):1255–68.
332. Thompson JA. New NCCN guidelines: recognition and management of immunotherapy-related toxicity. J Natl Compr Canc Netw. 2018;16(5s):594–6.
333. Eisen D, Griffiths CE, Ellis CN, Nickoloff BJ, Voorhees JJ. Cyclosporin wash for oral lichen planus. Lancet. 1990;335(8688):535–6.
334. Bettencourt M. Oral Lichen Planus Treated With Apremilast. J Drugs Dermatol. 2016;15(8):1026–8.
335. Carbone M, Goss E, Carrozzo M, Castellano S, Conrotto D, Broccoletti R, et al. Systemic and topical corticosteroid treatment of oral lichen planus: a comparative study with long-term follow-up. J Oral Pathol Med. 2003;32(6):323–9.
336. Passeron T, Lacour JP, Fontas E, Ortonne JP. Treatment of oral erosive lichen planus with 1% pimecrolimus cream: a double-blind, randomized, prospective trial with measurement of pimecrolimus levels in the blood. Arch Dermatol. 2007;143(4):472–6.
337. Malhotra AK, Khaitan BK, Sethuraman G, Sharma VK. Betamethasone oral mini-pulse therapy compared with topical triamcinolone acetonide (0.1%) paste in oral lichen planus: A randomized comparative study. J Am Acad Dermatol. 2008;58(4):596–602.
338. Lodi G, Manfredi M, Mercadante V, Murphy R, Carrozzo M. Interventions for treating oral lichen planus: corticosteroid therapies. Cochrane Database of Systematic Reviews. 2020(2).
339. Hodgson TA, Sahni N, Kaliakatsou F, Buchanan JA, Porter SR. Long-term efficacy and safety of topical tacrolimus in the management of ulcerative/erosive oral lichen planus. Eur J Dermatol. 2003;13(5):466–70.
340. da Silva EL, de Lima TB, Rados PV, Visioli F. Efficacy of topical non-steroidal immunomodulators in the treatment of oral lichen planus: a systematic review and meta-analysis. Clin Oral Investig. 2021;25(9):5149–69.
341. Byrd JA, Davis MD, Bruce AJ, Drage LA, Rogers RS 3rd. Response of oral lichen planus to topical tacrolimus in 37 patients. Arch Dermatol. 2004;140(12):1508–12.

342. Georgaki M, Piperi E, Theofilou VI, Pettas E, Stoufi E, Nikitakis NG. A randomized clinical trial of topical dexamethasone vs. cyclosporine treatment for oral lichen planus. Med Oral Patol Oral Cir Bucal. 2022;27(2):e113–24.
343. Monshi B, Ellersdorfer C, Edelmayer M, Dvorak G, Ganger C, Ulm C, et al. Topical Cyclosporine in Oral Lichen Planus-A Series of 21 Open-Label, Biphasic, Single-Patient Observations. J Clin Med. 2021;10(22):1–11.
344. Singh AR, Rai A, Aftab M, Jain S, Singh M. Efficacy of steroidal vs non-steroidal agents in oral lichen planus: a randomised, open-label study. J Laryngol Otol. 2017;131(1):69–76.
345. Platais C, Lalagianni N, Momen S, Ormond M, McParland H, Setterfield J. Efficacy of hydroxychloroquine in oral lichen planus: a retrospective review. Br J Dermatol. 2023;188(4):557–8.
346. Eisen D. Hydroxychloroquine sulfate (Plaquenil) improves oral lichen planus: An open trial. J Am Acad Dermatol. 1993;28(4):609–12.
347. Yeshurun A, Bergman R, Bathish N, Khamaysi Z. Hydroxychloroquine sulphate therapy of erosive oral lichen planus. Australas J Dermatol. 2019;60(2):e109–12.
348. Yeo L, Ormerod AD. Oral tacrolimus: a treatment option for recalcitrant erosive lichen planus. Clin Exp Dermatol. 2016;41(6):684–5.
349. Mahévas T, Bertinchamp R, Battistella M, Reygagne P, Oksenhendler E, Fieschi C, et al. Efficacy of oral sirolimus as salvage therapy in refractory lichen planus associated with immune deficiency. Br J Dermatol. 2018;179(3):771–3.
350. Dalmau J, Puig L, Roé E, Peramiquel L, Campos M, Alomar A. Successful treatment of oral erosive lichen planus with mycophenolate mofetil. J Eur Acad Dermatol Venereol. 2007;21(2):259–60.
351. Camisa C, Popovsky JL. Effective treatment of oral erosive lichen planus with thalidomide. Arch Dermatol. 2000;136(12):1442–3.
352. Arora P, Sardana K, Gautam RK, Malhotra P. Low dose thalidomide monotherapy for the treatment of recalcitrant erosive lichen planus. Dermatol Ther. 2019;32(2): e12824.
353. Katsue H, Higashi Y, Hatanaka M, Kanekura T. Patient with oral lichen planus successfully treated with irsogladine maleate. J Dermatol. 2020;47(2):e56–7.
354. Yarom N. Etanercept for the management of oral lichen planus. Am J Clin Dermatol. 2007;8(2):121.
355. Bender A, Fix C, Eubel V, Eming R, Pollmann R, Schmidt T, et al. Adjuvant high-dose intravenous immunoglobulins for recalcitrant erosive oral lichen planus: mixed clinical responses. Eur J Dermatol. 2018;28(4):496–501.
356. Hollis AN, Myers EL, Culton DA. A retrospective cohort study on the efficacy of metronidazole in oral lichen planus. Clin Exp Dermatol. 2023;48(12):1354–7.
357. Piñas L, Alkhraisat MH, Fernández RS, Anitua E. Biological therapy of refractory ulcerative oral lichen planus with plasma rich in growth factors. Am J Clin Dermatol. 2017;18(3):429–33.
358. Cosgarea R, Pollmann R, Sharif J, Schmidt T, Stein R, Bodea A, et al. Photodynamic therapy in oral lichen planus: a prospective case-controlled pilot study. Sci Rep. 2020;10(1):1667.
359. Hesse J, Schmalfuss A, Kvaal SI. Photodynamic therapy of oral lichen planus. Photochem Photobiol Sci. 2020;19(10):1271–9.
360. Zingoni A, Deboli T, Savoia P, Bernengo MG. Effectiveness of extracorporeal photochemotherapy in the treatment of a case of refractory erosive lichen planus. J Dermatolog Treat. 2010;21(2):119–21.
361. Guyot AD, Farhi D, Ingen-Housz-Oro S, Bussel A, Parquet N, Rabian C, et al. Treatment of refractory erosive oral lichen planus with extracorporeal photochemotherapy: 12 cases. Br J Dermatol. 2007;156(3):553–6.
362. Passeron T, Zakaria W, Ostovari N, Mantoux F, Lacour JP, Ortonne JP. Treatment of erosive oral lichen planus by the 308 nm excimer laser. Lasers Surg Med. 2004;34(3):205.
363. Khater MM, Khattab FM. Efficacy of 1064 Q switched Nd:YAG laser in the treatment of oral lichen planus. J Dermatolog Treat. 2020;31(6):655–9.
364. Lonsdale-Eccles AA, Velangi S. Topical pimecrolimus in the treatment of genital lichen planus: a prospective case series. Br J Dermatol. 2005;153(2):390–4.
365. Wee JS, Shirlaw PJ, Challacombe SJ, Setterfield JF. Efficacy of mycophenolate mofetil in severe mucocutaneous lichen planus: a retrospective review of 10 patients. Br J Dermatol. 2012;167(1):36–43.
366. Bradford J, Fischer G. Management of vulvovaginal lichen planus: a new approach. J Low Genit Tract Dis. 2013;17(1):28–32.
367. Poon F, De Cruz R, Hall A. Acitretin in erosive penile lichen planus. Australas J Dermatol. 2017;58(3):e87–90.
368. Vermeer HAB, Rashid H, Esajas MD, Oldhoff JM, Horváth B. The use of hydroxychloroquine as a systemic treatment in erosive lichen planus of the vulva and vagina. Br J Dermatol. 2021;185(1):201–3.
369. Lagerstedt M, Kotaniemi-Talonen L, Antonen J, Vaalasti A. Erosive vulvo-vaginal lichen planus treated with rituximab: a case report. Int J Gynaecol Obstet. 2022;156(1):172–3.
370. Wennberg AM. Vulvovaginal photodynamic therapy for genital erosive lichen planus. Br J Dermatol. 2015;173(5):1119–20.
371. Doche I, Romiti R, Hordinsky MK, Valente NS. "Normal-appearing" scalp areas are also affected in lichen planopilaris and frontal fibrosing alopecia: An observational histopathologic study of 40 patients. Exp Dermatol. 2020;29(3):278–81.
372. Lyakhovitsky A, Amichai B, Sizopoulou C, Barzilai A. A case series of 46 patients with lichen planopilaris: demographics, clinical evaluation, and treatment experience. J Dermatolog Treat. 2015;26(3):275–9.
373. Rácz E, Gho C, Moorman PW, Noordhoek Hegt V, Neumann HA. Treatment of frontal fibrosing alopecia and lichen planopilaris: a systematic review. J Eur Acad Dermatol Venereol. 2013;27(12):1461–70.
374. Strazzulla LC, Avila L, Lo Sicco K, Shapiro J. Novel treatment using low-dose naltrexone for lichen planopilaris. J Drugs Dermatol. 2017;16(11):1140–2.
375. Lee B, Elston DM. The uses of naltrexone in dermatologic conditions. J Am Acad Dermatol. 2019;80(6):1746–52.
376. Vañó-Galván S, Trindade de Carvalho L, Saceda-Corralo D, Rodrigues-Barata R, Kerkemeyer KL, Sinclair RD, et al. Oral minoxidil improves background hair thickness in lichen planopilaris. J Am Acad Dermatol. 2021;84(6):1684–6.
377. Peterson EL, Gutierrez D, Brinster NK, Lo Sicco KI, Shapiro J. Response of lichen planopilaris to pioglitazone hydrochloride. J Drugs Dermatol. 2019;18(12):1276–9.
378. Saber M, Bahraminejad M, Fatemi Naeini F, Mohaghegh F, Faghihi G, Hosseini SM. Comparison of systemic and topical isotretinoin in the treatment of facial lichen planopilaris: a randomized controlled trial. J Cosmet Dermatol. 2022;21(9):3896–904.
379. Babahosseini H, Tavakolpour S, Mahmoudi H, Balighi K, Teimourpour A, Ghodsi SZ, et al. Lichen planopilaris: retrospective study on the characteristics and treatment of 291 patients. J Dermatolog Treat. 2019;30(6):598–604.
380. Boyd AS, King LE Jr. Thalidomide-induced remission of lichen planopilaris. J Am Acad Dermatol. 2002;47(6):967–8.
381. Pathoulas JT, Flanagan KE, Su MY, Elmariah SB, Zhan Y, Walker CJ, et al. A prospective pilot study of narrowband

UV-B treatment of lichen planopilaris. J Am Acad Dermatol. 2022;87(3):703–5.
382. Fonda-Pascual P, Moreno-Arrones OM, Saceda-Corralo D, Rodrigues-Barata AR, Pindado-Ortega C, Boixeda P, et al. Effectiveness of low-level laser therapy in lichen planopilaris. J Am Acad Dermatol. 2018;78(5):1020–3.
383. Ujiie H, Shibaki A, Akiyama M, Shimizu H. Successful treatment of nail lichen planus with topical tacrolimus. Acta Derm Venereol. 2010;90(2):218–9.
384. Lipner SR. Nail lichen planus: a true nail emergency. J Am Acad Dermatol. 2019;80(6):e177–8.
385. Hwang JK, Grover C, Iorizzo M, Lebwohl MG, Piraccini BM, Rigopoulos DG, et al. Nail psoriasis and nail lichen planus: updates on diagnosis and management. J Am Acad Dermatol. 2024;90(3):585–96.
386. Florian B, Angelika J, Ernst SR. Successful treatment of palmoplantar nail lichen planus with cyclosporine. J Dtsch Dermatol Ges. 2014;12(8):724–5.
387. Irla N, Schneiter T, Haneke E, Yawalkar N. Nail lichen planus: successful treatment with etanercept. Case Rep Dermatol. 2010;2(3):173–6.
388. Pita da Veiga G, Pérez-Feal P, Moreiras-Arias N, Peteiro-García C, Suárez-Peñaranda JM, Vázquez-Veiga H, et al. Treatment of nail lichen planus with localized bath-PUVA. Photodermatol Photoimmunol Photomed. 2020;36(3):241–3.
389. Motamed-Sanaye A, Khazaee YF, Shokrgozar M, Alishahi M, Ahramiyanpour N, Amani M. JAK inhibitors in lichen planus: a review of pathogenesis and treatments. J Dermatolog Treat. 2022;33(8):3098–103.
390. Böll SL, Zahn CA, Schlapbach C. Rapid and sustained improvement of cutaneous lichen planus with oral JAK1 inhibitors. J Eur Acad Dermatol Venereol. 2024;38(1):e82–5.
391. Brumfiel CM, Patel MH, Severson KJ, Zhang N, Li X, Quillen JK, et al. Ruxolitinib cream in the treatment of cutaneous lichen planus: a prospective, open-label study. J Invest Dermatol. 2022;142(8):2109-16.e4.
392. Youssef S, Bordone LA. Oral tofacitinib effectively treating eruptive and hypertrophic cutaneous lichen planus. JAAD Case Rep. 2023;37:16–20.
393. Seiringer P, Lauffer F, Pilz AC, Boehmer D, Biedermann T, Eyerich K. Tofacitinib in hypertrophic lichen planus. Acta Derm Venereol. 2020;100(14):adv00220.
394. Solimani F, Mesas-Fernández A, Dilling A, Nast A, Hilke FJ, Ghoreschi FC, et al. The Janus kinase 1 inhibitor abrocitinib for the treatment of oral lichen planus. J Eur Acad Dermatol Venereol. 2023;37(8):e996–8.
395. Landells FM, Goudie S, McGrath J, Tibbo J, Landells ID. Successful treatment of erosive lichen planus with Upadacitinib complicated by oral squamous cell carcinoma. SAGE Open Med Case Rep. 2023;11:2050313x231213144.
396. Balestri R, Bortolotti R, Rech G, Girardelli CR, Zorzi MG, Magnano M. Treatment of oral erosive lichen planus with upadacitinib. JAMA Dermatol. 2022;158(4):457–8.
397. Kooybaran NR, Petzold G, Ströbel P, Schön MP, Mössner R. Alleviation of erosive oral and esophageal lichen planus by the JAK1 inhibitor upadacitinib. J Dtsch Dermatol Ges. 2021;19(12):1778–80.
398. Damsky W, Wang A, Olamiju B, Peterson D, Galan A, King B. Treatment of severe lichen planus with the JAK inhibitor tofacitinib. J Allergy Clin Immunol. 2020;145(6):1708-10.e2.
399. Kassels A, Edwards L, Kraus CN. Treatment of erosive vulvovaginal lichen planus with tofacitinib: a case series. JAAD Case Rep. 2023;40:14–8.
400. Kozlov M, Levit EK, Silvers DN, Brichkov I. Severe esophageal lichen planus treated with tofacitinib. Cutis. 2023;111(3):155–63.
401. He J, Weng T, Zhu W, Yang Y, Li C. Alleviation of isolated nail lichen planus by the JAK1/2 inhibitor Baricitinib: a case report. J Dermatolog Treat. 2023;34(1):2274816.
402. Chen LC, Ogbutor C, Kelley KJ, Senna MM. Topical tofacitinib for patients with lichen planopilaris and/or frontal fibrosing alopecia. J Am Acad Dermatol. 2024;90(6):1260–2.
403. Moussa A, Bhoyrul B, Asfour L, Kazmi A, Eisman S, Sinclair RD. Treatment of lichen planopilaris with baricitinib: a retrospective study. J Am Acad Dermatol. 2022;87(3):663–6.
404. Plante J, Eason C, Snyder A, Elston D. Tofacitinib in the treatment of lichen planopilaris: A retrospective review. J Am Acad Dermatol. 2020;83(5):1487–9.
405. Samuel C, Cornman H, Kambala A, Kwatra SG. A review on the safety of using JAK inhibitors in dermatology: clinical and laboratory monitoring. Dermatol Ther (Heidelb). 2023;13(3):729–49.
406. Paul J, Foss CE, Hirano SA, Cunningham TD, Pariser DM. An open-label pilot study of apremilast for the treatment of moderate to severe lichen planus: a case series. J Am Acad Dermatol. 2013;68(2):255–61.
407. Kim-Lim P, Thomas C. Crushed apremilast for the treatment of oral lichen planus. JAAD Case Rep. 2023;37:114–5.
408. Skullerud KH, Gjersvik P, Pripp AH, Qvigstad E, Helgesen ALO. Apremilast for genital erosive lichen planus in women (the AP-GELP Study): study protocol for a randomised placebo-controlled clinical trial. Trials. 2021;22(1):469.
409. Perschy L, Anzengruber F, Rappersberger K, Itzlinger-Monshi B, Aichelburg MC, Graf V, et al. Apremilast in oral lichen planus—a multicentric, retrospective study. J Dtsch Dermatol Ges. 2022;20(3):343–6.
410. Fage S, Johansen C. Severe and therapeutic challenging oral erosive lichen planus treated with oral roflumilast. Clin Exp Dermatol. 2023;48(5):556–7.
411. Viswanath V, Joshi P, Dhakne M, Dhoot D, Mahadkar N, Barkate H. Evaluation of the efficacy and safety of apremilast in the management of lichen planus. Clin Cosmet Investig Dermatol. 2022;15:2593–600.
412. Hafner J, Gubler C, Kaufmann K, Nobbe S, Navarini AA, French LE. Apremilast is effective in lichen planus mucosae-associated stenotic esophagitis. Case Rep Dermatol. 2016;8(2):224–6.
413. Mardani M, Mofidi H, Dastgheib L, Ranjbar S, Hamidizadeh N. Elevated serum interleukin-23 levels in patients with oral and cutaneous lichen planus. Mediators Inflamm. 2021;2021:5578568.
414. Lu R, Zeng X, Han Q, Lin M, Long L, Dan H, et al. Overexpression and selectively regulatory roles of IL-23/IL-17 axis in the lesions of oral lichen planus. Mediators Inflamm. 2014;2014: 701094.
415. Chen J, Feng J, Chen X, Xu H, Zhou Z, Shen X, et al. Immunoexpression of interleukin-22 and interleukin-23 in oral and cutaneous lichen planus lesions: a preliminary study. Mediators Inflamm. 2013;2013: 801974.
416. Pouralibaba F, Babaloo Z, Pakdel F, Aghazadeh M. Serum level of interleukin 17 in patients with erosive and non erosive oral lichen planus. J Dent Res Dent Clin Dent Prospects. 2013;7(2):91–4.
417. Husein-ElAhmed H, Steinhoff M. Potential role of INTERLEUKIN-17 in the pathogenesis of oral lichen planus: a systematic review with META-analysis. J Eur Acad Dermatol Venereol. 2022;36(10):1735–44.
418. Ismail FF, Sinclair R. Clinical healing of erosive oral lichen planus with tildrakizumab implicates the interleukin-23/interleukin-17 pathway in the pathogenesis of lichen planus. Australas J Dermatol. 2020;61(2):e244–5.
419. Maurelli M, Colato C, Tessari G, Girolomoni G. Lichen planopilaris coexisting with plaque psoriasis effectively treated with brodalumab. Dermatol Ther. 2020;33(6): e13967.

420. Manolache L, Seceleanu-Petrescu D, Benea V. Lichen planus patients and stressful events. J Eur Acad Dermatol Venereol. 2008;22(4):437–41.
421. Hong S, Fan R, Cohen JM. Lichen planus is associated with depression and anxiety: a cross-sectional study in the All of Us research program. Arch Dermatol Res. 2023;315(5):1417–9.
422. Morken CM, Bhullar PK, Boudreaux BW, Xie F, Zhang N, Lehman JS, et al. Availability of a validated cutaneous lichen planus disease index: a systematic review. Int J Dermatol. 2023;62(8):e461–3.
423. Xie F, Morken CM, Zhang N, Pittelkow MR, Sartori Valinotti JC, Comfere NI, et al. Development of the lichen planus quality of life questionnaire (LPQoL) informed by expert clinician input and patient feedback: a retrospective survey study. Arch Dermatol Res. 2023;315(6):1561–9.

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